

# Probability

## Contents

|                                                            |           |
|------------------------------------------------------------|-----------|
| Configuring R . . . . .                                    | 1         |
| <b>1 Core properties of probabilities</b>                  | <b>2</b>  |
| 1.1 Defining probabilities . . . . .                       | 2         |
| 1.2 Conditional probability . . . . .                      | 3         |
| <b>2 Key probability distributions</b>                     | <b>5</b>  |
| 2.1 The Bernoulli distribution . . . . .                   | 6         |
| 2.2 The Poisson distribution . . . . .                     | 7         |
| 2.3 The Negative-Binomial distribution . . . . .           | 14        |
| 2.4 Weibull Distribution . . . . .                         | 15        |
| <b>3 Characteristics of probability distributions</b>      | <b>15</b> |
| 3.1 Probability density function . . . . .                 | 15        |
| 3.2 Hazard function . . . . .                              | 16        |
| 3.3 Expectation . . . . .                                  | 16        |
| 3.3.1 Conditional distributions and expectations . . . . . | 20        |
| 3.4 Fubini–Tonelli for expectations . . . . .              | 28        |
| 3.5 Deviation, error, and noise . . . . .                  | 42        |
| 3.6 Variance and related characteristics . . . . .         | 42        |
| 3.7 The Central Limit Theorem . . . . .                    | 48        |
| <b>4 Additional resources</b>                              | <b>50</b> |
| <b>References</b>                                          | <b>50</b> |

---

## Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
```

```
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

---

Most of the content in this chapter should be review from UC Davis Epi 202.

## 1 Core properties of probabilities

### 1.1 Defining probabilities

**Definition 1.1** (Probability measure). A **probability measure**, often denoted  $\Pr()$  or  $P()$ , is a function whose domain is a  $\sigma$ -algebra<sup>a</sup> of possible outcomes,  $\mathcal{S}$ , and which satisfies the following properties:

1. For any statistical event  $A \in \mathcal{S}$ ,  $\Pr(A) \geq 0$ .
2. The probability of the union of all outcomes ( $\Omega \stackrel{\text{def}}{=} \cup \mathcal{S}$ ) is 1:

$$\Pr(\Omega) = 1$$

3. The probability of the union of countably many mutually disjoint events  $A_1, A_2, \dots$  (where  $A_i \cap A_j = \emptyset$  for all  $i \neq j$ ) is equal to the sum of their probabilities (*countable additivity* or *sigma-additivity*):

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

<sup>a</sup><https://en.wikipedia.org/wiki/%CE%A3-algebra>

Property 3 (*countable additivity*) is stronger than *finite additivity*, which only requires

$$\Pr(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \Pr(A_i)$$

for every finite collection of mutually disjoint events. Countable additivity implies finite additivity (set  $A_{n+1} = A_{n+2} = \dots = \emptyset$  in property 3, using  $\Pr(\emptyset) = 0$ ), but not vice versa: there exist set functions that satisfy finite additivity but fail countable additivity (see Wikipedia: Sigma-additive set function — An additive function which is not  $\sigma$ -additive<sup>1</sup>). Requiring countable additivity enables results such as the continuity of probability (if  $A_1 \supseteq A_2 \supseteq \dots$  with  $\bigcap_i A_i = \emptyset$ , then  $\Pr(A_i) \rightarrow 0$ ) and underpins the Theorem 1.4 for countable partitions.

**Theorem 1.1** (Probability of a subset's intersection). *If  $A$  and  $B$  are statistical events and  $A \subseteq B$ , then  $\Pr(A \cap B) = \Pr(A)$ .*

**i** Proof

*Proof.* Left to the reader for now. □

**Theorem 1.2** (An event and its complement sum to 1).

$$\Pr(A) + \Pr(\neg A) = 1$$

**i** Proof

*Proof.* By properties 2 and 3 of Definition 1.1. □

**Corollary 1.1** (Complement rule).

$$\Pr(\neg A) = 1 - \Pr(A)$$

**i** Proof

*Proof.* By Theorem 1.2 and algebra. □

<sup>1</sup>[https://en.wikipedia.org/wiki/Sigma-additive\\_set\\_function#An\\_additive\\_function\\_which\\_is\\_not\\_%CF%83-additive](https://en.wikipedia.org/wiki/Sigma-additive_set_function#An_additive_function_which_is_not_%CF%83-additive)

**Corollary 1.2** (Complement rule in probability ( $\pi$ ) notation). *If the probability of an outcome  $A$  is  $\Pr(A) = \pi$ , then the probability that  $A$  does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

**i** Proof

*Proof.* Using Corollary 1.1:

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) \\ &= 1 - \pi\end{aligned}$$

□

## 1.2 Conditional probability

**Definition 1.2** (Conditional probability). For two events  $A$  and  $B$  with  $\Pr(B) > 0$ , the **conditional probability** of  $A$  given  $B$ , denoted  $\Pr(A \mid B)$ , is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

**Theorem 1.3** (Law of conditional probability). *For any two events  $A$  and  $B$  with  $\Pr(B) > 0$ :*

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

**i** Proof

*Proof.* Rearranging Definition 1.2:

$$\begin{aligned}\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ \Pr(A \cap B) &= \Pr(A \mid B) \cdot \Pr(B) && \text{(multiply both sides by } \Pr(B))\end{aligned}$$

□

Exm

**Example 1.1** (Applying the law of conditional probability). Suppose 30% of adults exercise regularly ( $\Pr(E) = 0.30$ ), and among adults who exercise regularly, 60% have low blood pressure ( $\Pr(L \mid E) = 0.60$ ).

Then the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned}\Pr(L \cap E) &= \Pr(L \mid E) \cdot \Pr(E) \\ &= 0.60 \cdot 0.30 \\ &= 0.18\end{aligned}$$

**Theorem 1.4** (Law of total probability). *If  $B_1, B_2, \dots$  is a countable partition of the sample space (i.e., countably many mutually exclusive events whose union is the entire sample space),*

then for any event  $A$ :

$$\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i)$$

### i Proof

*Proof.* Since  $B_1, B_2, \dots$  partition the sample space, the events  $A \cap B_1, A \cap B_2, \dots$  are mutually exclusive and their union is  $A$ . By property 3 of Definition 1.1 (countable additivity), and then by Theorem 1.3:

$$\begin{aligned} \Pr(A) &= \sum_{i=1}^{\infty} \Pr(A \cap B_i) && \text{(countable additivity for partition of } A) \\ &= \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i) && \text{(law of conditional probability)} \end{aligned}$$

□

**Theorem 1.5** (Bayes' theorem). *For any two events  $A$  and  $B$  with  $\Pr(A) > 0$  and  $\Pr(B) > 0$ :*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)}$$

### i Proof

*Proof.* Apply Definition 1.2 to both  $\Pr(A \mid B)$  and  $\Pr(B \mid A)$ :

$$\begin{aligned} \Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} \\ &= \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)} \end{aligned}$$

The second equality follows from Theorem 1.3 applied to  $\Pr(B \cap A) = \Pr(B \mid A) \cdot \Pr(A)$ . □

### Exm

**Example 1.2** (Positive predictive value of a medical test). Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%. Let  $D$  be the event “person has the disease” and  $+$  be the event “test is positive”. Then:

- $\Pr(+ \mid D) = 0.99$  (sensitivity)
- $\Pr(\neg+ \mid \neg D) = 0.99$  (specificity), so the false positive rate is  $\Pr(+ \mid \neg D) = 1 - 0.99 = 0.01$
- $\Pr(D) = 0.07$  (prevalence)

By Bayes' theorem (Theorem 1.5) and the law of total probability (Theorem 1.4):

$$\begin{aligned}
\Pr(D \mid +) &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+)} \\
&= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+ \mid D) \cdot \Pr(D) + \Pr(+ \mid \neg D) \cdot \Pr(\neg D)} \\
&= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} \\
&= \frac{0.0693}{0.0693 + 0.0093} \\
&= \frac{0.0693}{0.0786} \\
&\approx 0.88
\end{aligned}$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).

## 2 Key probability distributions

Some distributions are typically used for outcome models (Table 1); other distributions are typically used for test statistics (Table 2).

Table 1: Distributions typically used for outcome models

| Distribution      | Uses                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------|
| Bernoulli         | Binary outcomes                                                                               |
| Binomial          | Sums of Bernoulli outcomes                                                                    |
| Poisson           | unbounded count outcomes                                                                      |
| Geometric         | Counts of non-events before an event occurs                                                   |
| Negative binomial | Mixtures of Poisson distributions, counts of non-events until a given number of events occurs |
| Normal (Gaussian) | Continuous outcomes without a more specific distribution                                      |
| exponential       | Time to event outcomes                                                                        |
| Gamma             | Time to event outcomes                                                                        |
| Weibull           | Time to event outcomes                                                                        |
| Log-normal        | Time to event outcomes                                                                        |

Table 2: Distributions typically used for test statistics

| Distribution          | Uses                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------|
| $\chi^2$              | Regression comparisons (asymptotic),<br>contingency table<br>independence tests,<br>goodness-of-fit tests |
| $F$                   | Gaussian model<br>comparisons (exact)                                                                     |
| $Z$ (standard normal) | Proportions, means,<br>regression coefficients<br>(asymptotic)                                            |
| $T$                   | Means, regression<br>coefficients in Gaussian<br>outcome models (exact)                                   |

## 2.1 The Bernoulli distribution

**Definition 2.1** (Bernoulli distribution). The **Bernoulli distribution** family for a random variable  $X$  is defined as:

$$\begin{aligned}\Pr(X = x) &\stackrel{\text{def}}{=} 1_{x \in \{0,1\}} \pi^x (1 - \pi)^{1-x} \\ &= \begin{cases} \pi, & x = 1 \\ 1 - \pi, & x = 0 \end{cases}\end{aligned}$$

## 2.2 The Poisson distribution



(a) Siméon Denis Poisson



(b) Les Poissons<sup>a</sup>

<sup>a</sup><https://youtu.be/UoJxBEQRLd0?t=12>

Figure 1: “Les Poissons”

**Exercise 2.1.** Define the Poisson distribution.

Solution

*Solution 2.1.*

**Def**

*Definition 2.2* (Poisson distribution).

$$P(Y = y) \stackrel{\text{def}}{=} \frac{\mu^y e^{-\mu}}{y!}, \quad y \in \mathbb{N} \quad (1)$$

(see Figure 2)

**Exercise 2.2.** What is the range of possible values for a Poisson distribution?

Solution

*Solution 2.2.*

$$\mathcal{R}(Y) \stackrel{\text{def}}{=} \{0, 1, 2, \dots\} = \mathbb{N}$$

**Theorem 2.1** (CDF of Poisson distribution).

$$P(Y \leq y) = e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} \quad (2)$$

### i Proof

*Proof.* For any  $y \geq 0$ , the event  $\{Y \leq y\}$  is the disjoint union of events  $\{Y = j\}$  for all non-negative integers  $j \leq \lfloor y \rfloor$ . Applying the Poisson PMF (Equation 1) and factoring out the common term  $e^{-\mu}$ :

$$\begin{aligned} P(Y \leq y) &= \sum_{j=0}^{\lfloor y \rfloor} P(Y = j) && \text{(disjoint union of events } Y = j) \\ &= \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j e^{-\mu}}{j!} && \text{(definition of Poisson PMF)} \\ &= e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} && \text{(factoring out } e^{-\mu} \text{ constant wrt } j) \end{aligned}$$

□

### Exm

**Example 2.1** (Example: Computing Poisson cumulative probabilities). For a Poisson random variable  $X \sim \text{Pois}(\mu = 2)$ , the probability of observing at most 2 events is computed as:

$$\begin{aligned} P(X \leq 2) &= e^{-2} \sum_{j=0}^2 \frac{2^j}{j!} && \text{(apply CDF formula with } \mu = 2, y = 2) \\ &= e^{-2} \left( \frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right) && \text{(expand terms for } j = 0, 1, 2) \\ &= e^{-2} (1 + 2 + 2) && \text{(simplify factorials and powers)} \\ &= 5e^{-2} \approx 0.677 && \text{(evaluate numerical value)} \end{aligned}$$

(see Figure 3)

```
library(dplyr)
pois_dists <- tibble(
  mu = c(0.5, 1, 2, 5, 10, 20)
) |>
  reframe(
    .by = mu,
    x = 0:30
  ) |>
  mutate(
    `P(X = x)` = dpois(x, lambda = mu),
    `P(X <= x)` = ppois(x, lambda = mu),
    mu = factor(mu)
  )

library(ggplot2)
library(latex2exp)

plot0 <- pois_dists |>
  ggplot(
    aes(
      x = x,
      y = `P(X = x)`,
      fill = mu,
      col = mu
    )
  ) +
```

```

theme(legend.position = "bottom") +
labs(
  fill = latex2exp::TeX("$\\mu$"),
  col = latex2exp::TeX("$\\mu$"),
  y = latex2exp::TeX("$\\Pr_{\\mu}(X = x)$")
)

plot1 <- plot0 +
  geom_segment(yend = 0) +
  facet_wrap(~mu)

print(plot1)

```

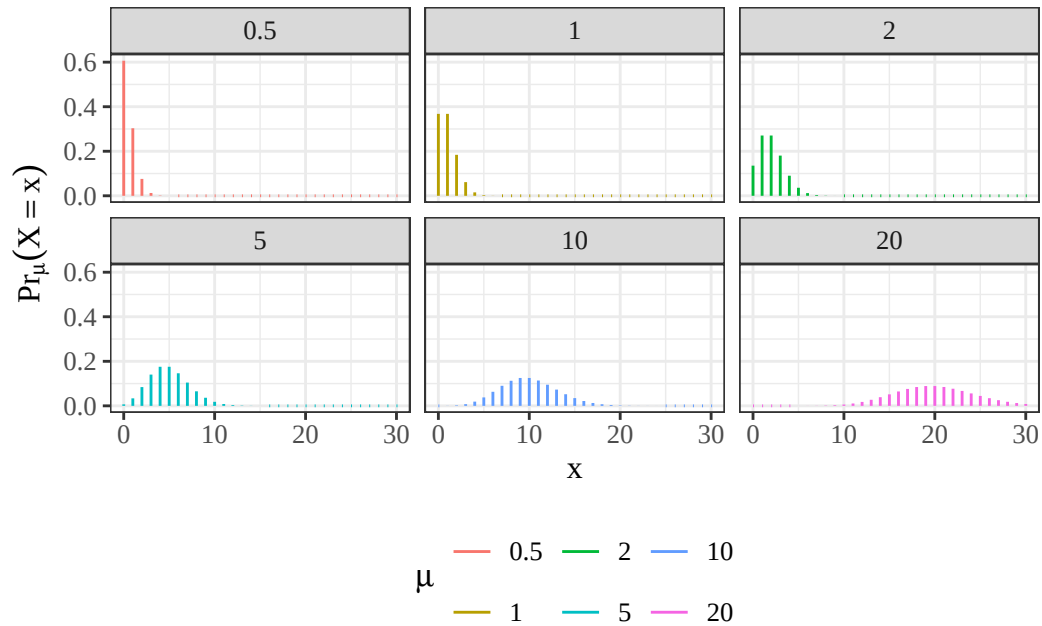


Figure 2: Poisson PMFs, by mean parameter  $\mu$

```

library(ggplot2)

plot2 <-
  plot0 +
  geom_step(alpha = 0.75) +
  aes(y = `P(X <= x)`) +
  labs(y = latex2exp::TeX("$\\Pr_{\\mu}(X \\leq x)$"))

print(plot2)

```

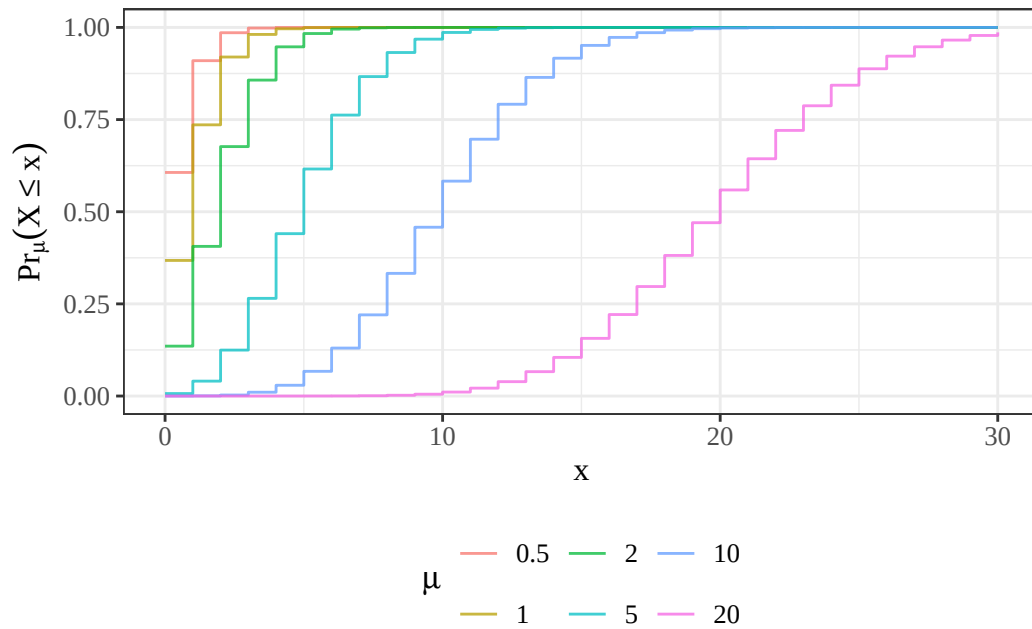


Figure 3: Poisson CDFs

**Exercise 2.3** (Poisson distribution functions). Let  $X \sim \text{Pois}(\mu = 3.75)$ .

Compute:

- $P(X = 4 | \mu = 3.75)$
- $P(X \leq 7 | \mu = 3.75)$
- $P(X > 5 | \mu = 3.75)$

Solution

*Solution.*

- $P(X = 4) = 0.19378$
- $P(X \leq 7) = 0.962379$
- $P(X > 5) = 0.177117$

**Theorem 2.2** (Properties of the Poisson distribution). *If  $X \sim \text{Pois}(\mu)$ , then:*

- $E[X] = \mu$
- $\text{Var}(X) = \mu$
- $P(X = x) = \frac{\mu}{x} P(X = x - 1)$
- *For  $x < \mu$ ,  $P(X = x) > P(X = x - 1)$*
- *For  $x = \mu$ ,  $P(X = x) = P(X = x - 1)$*
- *For  $x > \mu$ ,  $P(X = x) < P(X = x - 1)$*
- $\arg \max_x P(X = x) = \lfloor \mu \rfloor$

**Exercise 2.4.** Prove Theorem 2.2.

## Solution

*Solution.*

$$\begin{aligned}
 E[X] &= \sum_{x=0}^{\infty} x \cdot P(X = x) && \text{(definition of expected value)} \\
 &= 0 \cdot P(X = 0) + \sum_{x=1}^{\infty} x \cdot P(X = x) && \text{(separate } x = 0 \text{ term)} \\
 &= \sum_{x=1}^{\infty} x \cdot \frac{\mu^x e^{-\mu}}{x!} && \text{(substitute Poisson PMF)} \\
 &= \sum_{x=1}^{\infty} x \cdot \frac{\mu^x e^{-\mu}}{x \cdot (x-1)!} && \text{(definition of factorial } x!) \\
 &= \sum_{x=1}^{\infty} \frac{\mu^x e^{-\mu}}{(x-1)!} && \text{(cancel factor of } x) \\
 &= \mu \cdot \sum_{x=1}^{\infty} \frac{\mu^{x-1} e^{-\mu}}{(x-1)!} && \text{(factor out one power of } \mu) \\
 &= \mu \cdot \sum_{y=0}^{\infty} \frac{\mu^y e^{-\mu}}{y!} && \text{(change index variable } y \stackrel{\text{def}}{=} x-1) \\
 &= \mu \cdot 1 && \text{(PMF sums to 1 over state space)} \\
 &= \mu && \text{(simplify)}
 \end{aligned}$$

See also <https://statproofbook.github.io/P/poiss-mean>.

For the variance, see <https://statproofbook.github.io/P/poiss-var>.

## Accounting for exposure

**Definition 2.3** (Exposure magnitude). For many count outcomes, there is some sense of an **exposure magnitude**, such as **population size**, or **duration of observation**, which multiplicatively rescales the expected (mean) count.

**Exercise 2.5.** What are some examples of exposure magnitudes?

## Solution

*Solution.*

Table 3: Examples of exposure units

| outcome            | exposure units                              |
|--------------------|---------------------------------------------|
| disease incidence  | number of individuals exposed; time at risk |
| car accidents      | miles driven                                |
| worksite accidents | person-hours worked                         |
| population size    | size of habitat                             |

Exposure units are similar to the number of trials in a binomial distribution, but **in non-binomial count outcomes, there can be more than one event per unit of exposure**.

We can use  $t$  to represent continuous-valued exposures/observation durations, and  $n$  to represent discrete-valued exposures.

**Definition 2.4** (Event rate).

For a count outcome  $Y$  with exposure magnitude  $t$ , the **event rate** (denoted  $\lambda$ ) is defined as the mean of  $Y$  divided by the exposure magnitude. This mathematical relationship between mean and exposure magnitude is:

$$\begin{aligned}\mu &\stackrel{\text{def}}{=} \text{E}[Y|T = t] \\ \lambda &\stackrel{\text{def}}{=} \frac{\mu}{t}\end{aligned}\tag{3}$$

Event rate is somewhat analogous to odds in binary outcome models; it typically serves as an intermediate transformation between the mean of the outcome and the linear component of the model. However, in contrast with the odds function, the transformation  $\lambda = \mu/t$  is *not* considered part of the Poisson model's link function, and it treats the exposure magnitude covariate differently from the other covariates.

**Theorem 2.3** (Transformation function from event rate to mean). *For a count variable with mean  $\mu$ , event rate  $\lambda$ , and exposure magnitude  $t$ :*

$$\mu = \lambda \cdot t\tag{4}$$

**i** Proof

*Proof.*

$$\begin{aligned}\lambda &\stackrel{\text{def}}{=} \frac{\mu}{t} && \text{(definition of event rate @eq-def-event-rate)} \\ \mu &= \lambda \cdot t && \text{(multiply both sides by } t > 0\text{)}\end{aligned}$$

□

## Exm

**Example 2.2** (Example: Calculating expected counts from event rates). Suppose a city records a disease event rate of  $\lambda = 0.05$  cases per person-year. For a subpopulation with an exposure magnitude of  $t = 100$  person-years, the expected count of cases is:

$$\begin{aligned}\mu &= \lambda \cdot t && \text{(apply transformation formula @eq-lambda-to-mu)} \\ &= 0.05 \times 100 && \text{(substitute } \lambda = 0.05 \text{ and } t = 100\text{)} \\ &= 5 \text{ cases} && \text{(evaluate expected count)}\end{aligned}$$

## Solution

*Solution.* Start from definition of event rate and use algebra to solve for  $\mu$ .

Equation 4 is analogous to the inverse-odds function for binary variables.

**Theorem 2.4** (No exposure means no expected events). *When the exposure magnitude is 0, there is no opportunity for events to occur:*

$$\text{E}[Y|T = 0] = 0$$

### **i** Proof

*Proof.*

$$\begin{aligned} E[Y \mid T = 0] &= \lambda \cdot 0 && \text{(apply transformation @eq-lambda-to-mu at } t = 0\text{)} \\ &= 0 && \text{(multiplication by zero)} \end{aligned}$$

□

### Exm

**Example 2.3** (Example: Zero exposure time). If a subject is observed for  $t = 0$  person-years, no follow-up time has elapsed, so the expected number of incident events is  $E[Y \mid T = 0] = 0$ .

### **!** Important

The exposure magnitude,  $T$ , is *similar* to a covariate in linear or logistic regression. However, there is an important difference: in count regression, **there is no intercept corresponding to**  $E[Y \mid T = 0]$ . In other words, this model assumes that if there is no exposure, there can't be any events.

**Theorem 2.5** (Exposure is additive on the log scale). *If  $\mu = \lambda \cdot t$ , then:*

$$\log \mu = \log \lambda + \log t$$

### **i** Proof

*Proof.*

$$\begin{aligned} \log \mu &= \log(\lambda \cdot t) && \text{(substitute } \mu = \lambda \cdot t \text{ from @eq-lambda-to-mu)} \\ &= \log \lambda + \log t && \text{(logarithm product rule)} \end{aligned}$$

□

### Exm

**Example 2.4** (Example: Log-linear representation of expected counts). If a clinic sees an event rate of  $\lambda = 0.02$  events/day and  $t = 30$  days of observation:

$$\begin{aligned} \log \mu &= \log(0.02) + \log(30) && \text{(apply log-scale formula)} \\ &= -3.912 + 3.401 && \text{(evaluate natural logarithms)} \\ &= -0.511 && \text{(sum terms)} \end{aligned}$$

Exponentiating yields  $\mu = \exp\{-0.511\} \approx 0.60$  expected events.

**Definition 2.5** (Offset). When the linear component of a model involves a term without an unknown coefficient, that term is called an **offset**.

**Theorem 2.6** (Sum of independent Poisson random variables). *If  $X$  and  $Y$  are independent Poisson random variables with means  $\mu_X$  and  $\mu_Y$ , their sum,  $Z = X + Y$ , is also a Poisson random variable, with mean  $\mu_Z = \mu_X + \mu_Y$ .*

### i Proof

*Proof.* Using the probability-generating function or PMF convolution for independent non-negative integer random variables:

$$\begin{aligned} P(Z = z) &= \sum_{k=0}^z P(X = k) P(Y = z - k) && \text{(independence and convolution formula)} \\ &= \sum_{k=0}^z \frac{\mu_X^k e^{-\mu_X}}{k!} \frac{\mu_Y^{z-k} e^{-\mu_Y}}{(z-k)!} && \text{(substitute Poisson PMFs)} \\ &= \frac{e^{-(\mu_X + \mu_Y)}}{z!} \sum_{k=0}^z \frac{z!}{k!(z-k)!} \mu_X^k \mu_Y^{z-k} && \text{(factor out } e^{-(\mu_X + \mu_Y)}/z! \text{ and multiply by } z!/z!) \\ &= \frac{e^{-(\mu_X + \mu_Y)}}{z!} (\mu_X + \mu_Y)^z && \text{(binomial theorem)} \end{aligned}$$

This probability expression matches the PMF of a  $\text{Pois}(\mu_X + \mu_Y)$  random variable (see also [https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12\\_independent\\_rvs.pdf](https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12_independent_rvs.pdf), Example 3).  $\square$

### Exm

**Example 2.5** (Example: Aggregating independent region counts). Suppose Region A records  $X \sim \text{Pois}(\mu_X = 12)$  cases and Region B records  $Y \sim \text{Pois}(\mu_Y = 18)$  cases independently. The combined total count  $Z = X + Y$  follows a Poisson distribution:

$$\begin{aligned} Z &\sim \text{Pois}(\mu_X + \mu_Y) && \text{(apply sum theorem @thm - sum - pois)} \\ &= \text{Pois}(12 + 18) && \text{(substitute region means)} \\ &= \text{Pois}(30) && \text{(evaluate sum)} \end{aligned}$$

## 2.3 The Negative-Binomial distribution

**Definition 2.6** (Negative binomial distribution).

$$P(Y = y) \stackrel{\text{def}}{=} \frac{\mu^y}{y!} \cdot \frac{\Gamma(\rho + y)}{\Gamma(\rho) \cdot (\rho + \mu)^y} \cdot \left(1 + \frac{\mu}{\rho}\right)^{-\rho}$$

where  $\rho$  is an overdispersion parameter and  $\Gamma(x) = (x-1)!$  for integers  $x$ .

You don't need to memorize or understand this expression.

As  $\rho \rightarrow \infty$ , the second term converges to 1 and the third term converges to  $\exp\{-\mu\}$ , which brings us back to the Poisson distribution.

**Theorem 2.7** (Mean and variance of the negative binomial distribution). *If  $Y \sim \text{NegBin}(\mu, \rho)$ , then:*

- $E[Y] = \mu$
- $\text{Var}(Y) = \mu + \frac{\mu^2}{\rho} > \mu$

## 2.4 Weibull Distribution

$$\begin{aligned}p(t) &= \alpha \lambda x^{\alpha-1} e^{-\lambda x^\alpha} \\ \lambda(t) &= \alpha \lambda x^{\alpha-1} \\ S(t) &= e^{-\lambda x^\alpha} \\ E(T) &= \Gamma(1 + 1/\alpha) \cdot \lambda^{-1/\alpha}\end{aligned}$$

When  $\alpha = 1$ , the Weibull distribution reduces to the exponential distribution. When  $\alpha > 1$  the hazard is increasing and when  $\alpha < 1$  the hazard is decreasing. This property provides more flexibility than the exponential.

We will see more of this distribution later.

## 3 Characteristics of probability distributions

### 3.1 Probability density function

**Definition 3.1** (probability density). If  $X$  is a continuous random variable, then the **probability density** of  $X$  at value  $x$ , denoted  $f(x)$ ,  $f_X(x)$ ,  $p(x)$ ,  $p_X(x)$ , or  $p(X = x)$ , is defined as the limit of the **probability** (mass) that  $X$  is in an interval around  $x$ , divided by the width of that interval, as that width reduces to 0.

$$f(x) \stackrel{\text{def}}{=} \lim_{\Delta \rightarrow 0} \frac{P(X \in [x, x + \Delta])}{\Delta}$$

See also Rothman et al. (2021) (Chapter 22, p. 535) and [https://en.wikipedia.org/wiki/Probability\\_density\\_function#Formal\\_definition](https://en.wikipedia.org/wiki/Probability_density_function#Formal_definition)

**Definition 3.2** (Cumulative distribution function (CDF)). For a random variable  $X$ , its population CDF is

$$F(t) \stackrel{\text{def}}{=} \Pr(X \leq t), \quad t \in \mathbb{R}.$$

**Definition 3.3** (Quantile function (population inverse CDF)). For a random variable  $X$  with **cumulative distribution function (CDF)**  $F$ , its population quantile function (generalized inverse of  $F$ ) is

$$Q(p) \stackrel{\text{def}}{=} \inf\{t : F(t) \geq p\}, \quad 0 < p \leq 1.$$

**Theorem 3.1** (Density function is derivative of CDF). *The density function  $f(t)$  or  $p(T = t)$  for a random variable  $T$  at value  $t$  is equal to the derivative of the cumulative probability function  $F(t) \stackrel{\text{def}}{=} P(T \leq t)$ ; that is:*

$$f(t) \stackrel{\text{def}}{=} \frac{\partial}{\partial t} F(t)$$

**Theorem 3.2** (Density functions integrate to 1). *For any density function  $f(x)$ ,*

$$\int_{x \in \mathcal{R}(X)} f(x) dx = 1$$

## 3.2 Hazard function

**Definition 3.4** (Hazard function, hazard rate, hazard rate function).

The **hazard function**, **hazard rate**, **hazard rate function**, for a random variable  $T$  at value  $t$ , typically denoted as  $h(t)$  <sup>2</sup> or  $\lambda(t)$ , <sup>3</sup> is the conditional density<sup>a</sup> of  $T$  at  $t$ , given  $T \geq t$ . Specifically, the hazard function is defined as:

$$\lambda(t) \stackrel{\text{def}}{=} p(T = t | T \geq t)$$

If  $T$  represents the time at which an event occurs, then  $\lambda(t)$  is the probability that the event occurs at time  $t$ , given that it has not occurred prior to time  $t$ .

The name “hazard” carries a connotation that the event is undesirable — death, relapse, equipment failure, etc. When the event in question is neutral or desirable (recovery, conception, graduation, response to treatment), the same quantity  $\lambda(t)$  is often called the **event incidence rate** instead. This terminology is parallel to the convention that conditional probabilities of undesirable events are called **risks**, while the same conditional probabilities for neutral/desirable events are simply called **probabilities**. The math is identical; only the name changes with the valence of the event.

<sup>a</sup>[probability.qmd#def-pdf](#)

Table 4: Probability distribution functions

| Name                                   | Symbols            | Definition                         |
|----------------------------------------|--------------------|------------------------------------|
| Probability density function (PDF)     | $f(t), p(t)$       | $p(T = t)$                         |
| Cumulative distribution function (CDF) | $F(t), P(t)$       | $P(T \leq t)$                      |
| Survival function                      | $S(t), \bar{F}(t)$ | $P(T > t)$                         |
| Hazard function                        | $\lambda(t), h(t)$ | $p(T = t   T \geq t)$              |
| Cumulative hazard function             | $\Lambda(t), H(t)$ | $\int_{u=-\infty}^t \lambda(u) du$ |
| Log-hazard function                    | $\eta(t)$          | $\log\{\lambda(t)\}$               |

$$f(t) \xleftarrow{\frac{-S'(t)}{S(t)\lambda(t)}} S(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \xleftarrow{\int_{u=0}^t \lambda(u) du} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$f(t) \xrightarrow{\frac{f(t)/\lambda(t)}{\int_{u=t}^{\infty} f(u) du}} S(t) \xrightarrow{-\log S(t)} \Lambda(t) \xrightarrow{\Lambda'(t)} \lambda(t) \xrightarrow{\log\{\lambda(t)\}} \eta(t)$$

## 3.3 Expectation

**Definition 3.5** (Expectation, expected value, population mean ). The **expectation**, **expected value**, or **population mean** of a *continuous* random variable  $X$ , denoted  $E[X]$ ,  $\mu(X)$ , or  $\mu_X$ , is the weighted mean of  $X$ ’s possible values, weighted by the [probability density function](#) of those values:

$$E[X] \stackrel{\text{def}}{=} \int_{x \in \mathcal{R}(X)} x \cdot p(X = x) dx$$

The **expectation**, **expected value**, or **population mean** of a *discrete* random variable  $X$ , denoted  $E[X]$ ,  $\mu(X)$ , or  $\mu_X$ , is the mean of  $X$ ’s possible values, weighted by the probability mass function of those values:

<sup>3</sup>for example in Dobson and Barnett (2018), Vittinghoff et al. (2012), Klein and Moeschberger (2003), and Kleinbaum and Klein (2012)

<sup>3</sup>for example, in Rothman et al. (2021) and Kalbfleisch and Prentice (2011)

$$E[X] \stackrel{\text{def}}{=} \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x)$$

(c.f. [https://en.wikipedia.org/wiki/Expected\\_value](https://en.wikipedia.org/wiki/Expected_value))

**Theorem 3.3** (Expectation of the Bernoulli distribution). *The expectation of a Bernoulli random variable with parameter  $\pi$  is:*

$$E[X] = \pi$$

#### **i** Proof

*Proof.*

$$\begin{aligned} E[X] &= \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x) && \text{(definition of expectation for discrete r.v.)} \\ &= \sum_{x \in \{0,1\}} x \cdot P(X = x) && \text{(range of Bernoulli r.v. is } \{0,1\}) \\ &= (0 \cdot P(X = 0)) + (1 \cdot P(X = 1)) && \text{(expand sum over } x = 0 \text{ and } x = 1) \\ &= (0 \cdot (1 - \pi)) + (1 \cdot \pi) && \text{(substitute Bernoulli PMF values)} \\ &= 0 + \pi && \text{(multiplication by 0 and 1)} \\ &= \pi && \text{(addition of 0)} \end{aligned}$$

□

**Theorem 3.4** (Expectation of time-to-event variables). *If  $T$  is a non-negative random variable, then:*

$$\mu(T|\tilde{X} = \tilde{x}) = \int_{t=0}^{\infty} S(t) dt$$

#### **i** Proof

*Proof.* We prove the continuous case, in which  $T$  has a density  $f$ . The result follows from applying Tonelli's theorem (condition (a) of Fubini–Tonelli<sup>a</sup>) to the function  $g(t, u) = f(u) \cdot \mathbb{1}(0 \leq t \leq u)$  on the product space  $[0, \infty) \times [0, \infty)$ :  $g$  is nonnegative everywhere and vanishes outside the (unbounded) triangular region  $D = \{(t, u) : 0 \leq t \leq u < \infty\}$ , so the iterated integrals over  $D$  are exchangeable.

Since  $f(u) \geq 0$ , condition (a) of Fubini–Tonelli<sup>b</sup> (the nonnegative case, **Tonelli's theorem**) applies, and we may exchange the order of integration over  $D$ :

$$\begin{aligned}
E[T] &= \int_{u=0}^{\infty} u f(u) du \\
&= \int_{u=0}^{\infty} \left( \int_{t=0}^u 1 dt \right) f(u) du \\
&= \int_{u=0}^{\infty} \int_{t=0}^u f(u) dt du \\
&= \int_{t=0}^{\infty} \int_{u=t}^{\infty} f(u) du dt \\
&= \int_{t=0}^{\infty} P(T > t) dt \\
&= \int_{t=0}^{\infty} S(t) dt.
\end{aligned}$$

□

<sup>a</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)  
<sup>b</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)

Exm

**Example 3.1** (Mean of an exponential random variable via survival function). Let  $T \sim \text{Exponential}(\lambda)$ , so  $S(t) = e^{-\lambda t}$  for  $t \geq 0$ . By Theorem 3.4:

$$\begin{aligned}
E[T] &= \int_0^{\infty} S(t) dt \\
&= \int_0^{\infty} e^{-\lambda t} dt \\
&= \left[ -\frac{1}{\lambda} e^{-\lambda t} \right]_0^{\infty} \\
&= \frac{1}{\lambda},
\end{aligned}$$

confirming the standard result  $E[T] = 1/\lambda$ .

**Theorem 3.5** (Law of the Unconscious Statistician (LOTUS)). **Discrete case.** For any function  $g$  of a discrete random variable  $X$ :

$$E[g(X)] = \sum_{x \in \mathcal{R}(X)} g(x) \cdot P(X = x)$$

**Continuous case.** For any function  $g$  of a continuous random variable  $X$  with density  $p(X = x)$ :

$$E[g(X)] = \int_{x \in \mathcal{R}(X)} g(x) \cdot p(X = x) dx$$

**i** Proof

*Proof.* We prove the discrete case.

Let  $Y = g(X)$ . By Definition 3.5 applied to  $Y$ :

$$\begin{aligned}
\mathbb{E}[g(X)] &= \mathbb{E}[Y] && \text{(substitution } Y = g(X)) \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \mathbb{P}(Y = y) && \text{(definition of expectation for discrete r.v.)} \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \mathbb{P}(g(X) = y) && \text{(substitute } Y = g(X) \text{ in probability expression)} \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \sum_{\substack{x \in \mathcal{R}(X) \\ g(x) = y}} \mathbb{P}(X = x) && \text{(law of total probability over } \{x : g(x) = y\}) \\
&= \sum_{x \in \mathcal{R}(X)} g(x) \cdot \mathbb{P}(X = x) && \text{(rearrange double sum grouping by } g(x))
\end{aligned}$$

where the last equality follows by rearranging the double sum, grouping each term  $x$  by its image  $y = g(x)$ .

The continuous case is the density-weighted analogue of this argument, but a fully rigorous proof needs the general change-of-variables theorem for integrals against a pushforward measure — approximating  $g$  by simple functions and passing to the limit — which is beyond Epi 204's scope (Billingsley 1995; Gut 2013; Casella and Berger 2002; Wikipedia contributors 2026).  $\square$

---

LOTUS says that to compute  $\mathbb{E}[g(X)]$ , we do not need to first find the distribution of  $g(X)$ ; we can compute the expectation directly using the distribution of  $X$ .

---

Exm

**Example 3.2** (Expected value of  $X^2$  for a Bernoulli variable). Let  $X \sim \text{Ber}(\pi)$ . By LOTUS (Theorem 3.5, discrete case):

$$\begin{aligned}
\mathbb{E}[X^2] &= \sum_{x \in \{0,1\}} x^2 \cdot \mathbb{P}(X = x) \\
&= 0^2 \cdot \mathbb{P}(X = 0) + 1^2 \cdot \mathbb{P}(X = 1) \\
&= 0^2 \cdot (1 - \pi) + 1^2 \cdot \pi \\
&= 0 + \pi \\
&= \pi
\end{aligned}$$

Exm

**Example 3.3** (Expected value of  $X^2$  for a Uniform(0,1) variable). Let  $X \sim \text{Uniform}(0,1)$ , so  $p(X = x) = 1$  for  $x \in [0,1]$ . By LOTUS (Theorem 3.5, continuous case):

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_0^1 x^2 \cdot p(X = x) dx && \text{(LOTUS, continuous case)} \\
&= \int_0^1 x^2 \cdot 1 dx && \text{(p(X = x) = 1 on [0,1])} \\
&= \left[ \frac{x^3}{3} \right]_0^1 && \text{(antiderivative of } x^2) \\
&= \frac{1}{3}. && \text{(evaluate at the bounds)}
\end{aligned}$$


---

### 3.3.1 Conditional distributions and expectations

**Definition 3.6** (Conditional probability mass function). Let  $X$  and  $Y$  be jointly distributed discrete random variables. The **conditional probability mass function** of  $Y$  given  $X = x$  (for values of  $x$  with  $P(X = x) > 0$ ) is:

$$P(Y = y \mid X = x) \stackrel{\text{def}}{=} \frac{P(X = x, Y = y)}{P(X = x)}$$

Exm

**Example 3.4** (Conditional PMF from a vaccine trial). The `vaccine` dataset in the `dobson` package (Dobson and Barnett (2018), Table 9.6) records the response to treatment in a vaccine trial:  $X$  is treatment group (placebo or vaccine) and  $Y$  is response severity (small, moderate, or large).

```
library(dobson)
library(dplyr)
data(vaccine)
response_levels <- c("small", "moderate", "large")
vaccine_tab <- vaccine |>
  mutate(response = factor(response, levels = response_levels)) |>
  xtabs(frequency ~ treatment + response, data = _)
pander::pander(as.data.frame.matrix(vaccine_tab))
```

|         | small | moderate | large |
|---------|-------|----------|-------|
| placebo | 25    | 8        | 5     |
| vaccine | 6     | 18       | 11    |

The joint PMF of  $(X, Y)$  is the table of frequencies divided by the total  $n = 73$  trial participants. The marginal probability  $P(X = \text{placebo})$  is:

$$\begin{aligned} P(X = \text{placebo}) &= P(X = \text{placebo}, Y = \text{small}) + P(X = \text{placebo}, Y = \text{moderate}) + P(X = \text{placebo}, Y = \text{large}) \\ &= \frac{25}{73} + \frac{8}{73} + \frac{5}{73} \\ &= \frac{38}{73} \end{aligned}$$

By Definition 3.6, the conditional PMF of  $Y$  given  $X = \text{placebo}$  is:

$$\begin{aligned} P(Y = \text{small} \mid X = \text{placebo}) &= \frac{P(X = \text{placebo}, Y = \text{small})}{P(X = \text{placebo})} \\ &= \frac{25/73}{38/73} \\ &= \frac{25}{38} \end{aligned}$$

**Definition 3.7** (Conditional probability density function). Let  $X$  and  $Y$  be jointly distributed continuous random variables with joint density  $p(X = x, Y = y)$  and marginal density  $p(X = x)$ . The **conditional probability density function** of  $Y$  given  $X = x$  (for values of  $x$  with  $p(X = x) > 0$ ) is:

$$p(Y = y \mid X = x) \stackrel{\text{def}}{=} \frac{p(X = x, Y = y)}{p(X = x)}$$

Exm

**Example 3.5** (Conditional PDF from a bivariate normal model of birthweight data). The `birthweight` dataset in the `dobson` package (Dobson and Barnett (2018), Table 2.3) records gestational age (weeks) and birthweight (grams) for 12 boys and 12 girls. Let  $X$  be gestational age and  $Y$  be birthweight, pooling both sexes into  $n = 24$  observations.

```
library(dobson)
data(birthweight)
ga <- c(birthweight[["boys gestational age"]],
        birthweight[["girls gestational age"]])
wt <- c(birthweight[["boys weight"]], birthweight[["girls weight"]])
mu_x <- mean(ga)
sigma_x <- sd(ga)
mu_y <- mean(wt)
sigma_y <- sd(wt)
rho <- cor(ga, wt)
beta <- cov(ga, wt) / var(ga)
intercept <- mu_y - beta * mu_x
x0 <- 40
marg_dens_x0 <- dnorm(x0, mean = mu_x, sd = sigma_x)
cond_mean <- intercept + beta * x0
cond_sd <- sigma_y * sqrt(1 - rho^2)
pander::pander(data.frame(
  parameter = c("mu_x", "sigma_x", "mu_y", "sigma_y", "rho"),
  estimate = round(c(mu_x, sigma_x, mu_y, sigma_y, rho), 4)
))
```

| parameter | estimate |
|-----------|----------|
| mu_x      | 38.54    |
| sigma_x   | 1.817    |
| mu_y      | 2,968    |
| sigma_y   | 282.1    |
| rho       | 0.7443   |

Modeling  $(X, Y)$  as bivariate normal with parameters set equal to these sample moments, the joint density is (a standard result; e.g. Casella and Berger (2002)):

$$p(X = x, Y = y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[ \frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right]}$$

A further standard fact about the bivariate normal (Casella and Berger (2002)) is that the marginal distribution of  $X$  is  $X \sim N(\mu_X, \sigma_X^2)$ . At  $x = 40$  weeks (a full-term pregnancy),  $\mu_X = 38.5417$  and  $\sigma_X = 1.8173$ , so:

$$\begin{aligned} p(X = 40) &= \frac{1}{\sigma_X\sqrt{2\pi}} e^{-\frac{(40-\mu_X)^2}{2\sigma_X^2}} \\ &= \frac{1}{1.8173\sqrt{2\pi}} e^{-\frac{(40-38.5417)^2}{2(1.8173)^2}} \\ &\approx 0.1591 \end{aligned}$$

By Definition 3.7, dividing the joint density by this marginal density and simplifying the exponent (completing the square in  $y$ ; Casella and Berger (2002)) gives the conditional PDF of  $Y$  given  $X = 40$ , which is itself normal with mean shifted along the regression line and variance reduced by a factor of  $1 - \rho^2$ :

$$\begin{aligned} p(Y = y \mid X = 40) &= \frac{p(X = 40, Y = y)}{p(X = 40)} \\ &= \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)} \left[ \frac{(40-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(40-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right] + \frac{(40-\mu_X)^2}{2\sigma_X^2}} \\ &= \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho^2)}} e^{-\frac{\rho^2(40-\mu_X)^2}{2\sigma_X^2(1-\rho^2)} + \frac{\rho(40-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y(1-\rho^2)} - \frac{(y-\mu_Y)^2}{2\sigma_Y^2(1-\rho^2)}} \\ &= \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho^2)}} e^{-\frac{1}{2\sigma_Y^2(1-\rho^2)} \left[ \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} (40-\mu_X)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (40-\mu_X)(y-\mu_Y) + (y-\mu_Y)^2 \right]} \\ &= \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho^2)}} e^{-\frac{(y - [\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (40-\mu_X)])^2}{2\sigma_Y^2(1-\rho^2)}} \end{aligned}$$

The first equality substitutes the joint and marginal densities and combines their prefactors and exponents; the second equality combines the two  $(40 - \mu_X)^2$  terms using  $1 - \frac{1}{1-\rho^2} = \frac{-\rho^2}{1-\rho^2}$ ; the third equality factors  $-\frac{1}{2\sigma_Y^2(1-\rho^2)}$  out of the bracket (multiplying through confirms it reproduces the previous line); and the fourth equality completes the square in  $y$ , since expanding  $(y - [\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (40 - \mu_X)])^2$  reproduces exactly the bracketed quadratic in the third equality.

So  $Y \mid X = 40 \sim N(3136.15, 188.37^2)$ : the conditional mean, 3136.15 g, is exactly the fitted regression line's prediction at  $x = 40$  ( $-1484.9846 + 115.5283 \times 40 = 3136.15$ ), matching R's `lm(wt ~ ga)` fit directly:

```
pander::pander(coef(lm(wt ~ ga)))
```

| (Intercept) | ga    |
|-------------|-------|
| -1,485      | 115.5 |

**Definition 3.8** (Conditional expectation). **Discrete case.** Let  $X$  and  $Y$  be jointly distributed discrete random variables. The **conditional expectation** of  $Y$  given  $X = x$ , using Definition 3.6, is:

$$E[Y | X = x] \stackrel{\text{def}}{=} \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y | X = x)$$

**Continuous case.** Let  $X$  and  $Y$  be jointly distributed continuous random variables. The **conditional expectation** of  $Y$  given  $X = x$ , using Definition 3.7, is:

$$E[Y | X = x] \stackrel{\text{def}}{=} \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y | X = x) dy$$

Exm

**Example 3.6** (Conditional expectation from real trial and birthweight data). **Discrete case.** Continuing Example 3.4, score the vaccine trial's response levels small = 1, moderate = 2, large = 3. The conditional PMF of  $Y$  given  $X = \text{placebo}$  (from Example 3.4) is:

$$\begin{aligned} P(Y = \text{small} | X = \text{placebo}) &= \frac{25}{38} \\ P(Y = \text{moderate} | X = \text{placebo}) &= \frac{8}{38} \\ P(Y = \text{large} | X = \text{placebo}) &= \frac{5}{38} \end{aligned}$$

so:

$$\begin{aligned} E[Y | X = \text{placebo}] &= 1 \cdot \frac{25}{38} + 2 \cdot \frac{8}{38} + 3 \cdot \frac{5}{38} \\ &= \frac{25 + 16 + 15}{38} \\ &= \frac{56}{38} \\ &= \frac{28}{19} \\ &\approx 1.47 \end{aligned}$$

**Continuous case.** Continuing Example 3.5,  $Y | X = 40 \sim N(3136.15, 188.37^2)$ . Since the mean of a normal random variable is its own location parameter, integrating  $y$  against this conditional density directly gives:

$$\begin{aligned} E[Y | X = 40] &= \int_{-\infty}^{\infty} y \cdot p(Y = y | X = 40) dy \\ &= 3136.15 \text{ g} \end{aligned}$$

matching the fitted regression line's prediction at  $x = 40$  weeks exactly, as expected since the conditional mean of a bivariate normal *is* the linear regression of  $Y$  on  $X$ .

**Definition 3.9** (Conditional expectation: mixed case). Suppose exactly one of  $X, Y$  is discrete and the other is continuous. Write  $p(X = x, Y = y)$  for their **joint density-mass function**: a probability density in the continuous variable and a probability mass in the discrete variable.  **$X$  discrete,  $Y$  continuous.** Here  $p(X = x, Y = y)$  is, for each fixed  $x$ , a probability density in  $y$ , scaled so that  $\int_y p(X = x, Y = y) dy = P(X = x)$ . The conditional PDF of  $Y$  given  $X = x$  (for values of  $x$  with  $P(X = x) > 0$ ) is:

$$p(Y = y | X = x) \stackrel{\text{def}}{=} \frac{p(X = x, Y = y)}{P(X = x)}$$

and the conditional expectation of  $Y$  given  $X = x$  is:

$$E[Y | X = x] \stackrel{\text{def}}{=} \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y | X = x) dy$$

**$X$  continuous,  $Y$  discrete.** Here  $p(X = x, Y = y)$  is, for each fixed  $y$ , a probability density in  $x$ , scaled so that  $\sum_y p(X = x, Y = y) = p(X = x)$ . The conditional PMF of  $Y$  given  $X = x$  (for values of  $x$  with  $p(X = x) > 0$ ) is:

$$P(Y = y | X = x) \stackrel{\text{def}}{=} \frac{p(X = x, Y = y)}{p(X = x)}$$

and the conditional expectation of  $Y$  given  $X = x$  is:

$$E[Y | X = x] \stackrel{\text{def}}{=} \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y | X = x)$$

Exm

**Example 3.7** (Conditional expectation, one discrete variable and one continuous variable).  **$X$  discrete,  $Y$  continuous.** The `plasma` dataset in the `dobson` package (Dobson and Barnett (2018), Table 6.25) records plasma inorganic phosphate levels (mg/dL) one hour after a glucose tolerance test, for hyperinsulinemic obese (H-O) and control (C) participants. Let  $X$  be group and  $Y$  be phosphate level.

```
library(dobson)
library(dplyr)
data(plasma)
plasma_subset <- plasma |> filter(Group %in% c("H-O", "C"))
plasma_summary <- plasma_subset |>
  summarise(n = n(), mean = mean(phosphate), sd = sd(phosphate), .by = Group)
n_ho <- plasma_summary$n[plasma_summary$Group == "H-O"]
n_c <- plasma_summary$n[plasma_summary$Group == "C"]
mean_ho <- plasma_summary$mean[plasma_summary$Group == "H-O"]
sd_ho <- plasma_summary$sd[plasma_summary$Group == "H-O"]
mean_c <- plasma_summary$mean[plasma_summary$Group == "C"]
sd_c <- plasma_summary$sd[plasma_summary$Group == "C"]
pander::pander(plasma_summary)
```

| Group | n  | mean  | sd     |
|-------|----|-------|--------|
| H-O   | 11 | 3.945 | 0.7776 |
| C     | 12 | 2.783 | 0.4086 |

Modeling phosphate as approximately normal within each group, with parameters set equal to each group's sample mean and SD, and  $P(X = \text{H-O}) = 11/23$ :

$$p(Y = y \mid X = \text{H-O}) \approx N(3.95, 0.78^2)$$

$$p(Y = y \mid X = \text{C}) \approx N(2.78, 0.41^2)$$

By Definition 3.9, since the mean of a normal random variable is its own location parameter:

$$E[Y \mid X = \text{H-O}] = 3.95 \text{ mg/dL}, \quad E[Y \mid X = \text{C}] = 2.78 \text{ mg/dL}$$

**X continuous, Y discrete.** The `senility` dataset in the `dobson` package (Dobson and Barnett (2018), Table 7.8) records, for 54 elderly people, a WAIS (Wechsler Adult Intelligence Scale) score and whether symptoms of senility were present. Let  $X$  be WAIS score and  $Y$  indicate senility symptoms.

```
data(senility)
senility_fit <- glm(s ~ x, data = senility, family = binomial)
b0 <- coef(senility_fit)[["(Intercept)"]]
b1 <- coef(senility_fit)[["x"]]
pander::pander(coef(senility_fit))
```

| (Intercept) | x       |
|-------------|---------|
| 2.404       | -0.3235 |

Modeling  $P(Y = 1 \mid X = x)$  with the fitted logistic regression:

$$\text{logit } P(Y = 1 \mid X = x) = 2.4040 - 0.3235 x$$

By Definition 3.9, since  $Y \mid X = x$  is Bernoulli:

$$\begin{aligned} E[Y \mid X = x] &= 0 \cdot P(Y = 0 \mid X = x) + 1 \cdot P(Y = 1 \mid X = x) \\ &= P(Y = 1 \mid X = x) \\ &= \frac{1}{1 + e^{-(2.4040 - 0.3235 x)}} \end{aligned}$$

```
senility_p10 <- predict(senility_fit,
                        newdata = data.frame(x = 10), type = "response")
senility_p15 <- predict(senility_fit,
                        newdata = data.frame(x = 15), type = "response")
```

At  $x = 10$ :  $E[Y \mid X = 10] = 0.3$ . At  $x = 15$ :  $E[Y \mid X = 15] = 0.08$  — a lower WAIS score is associated with higher predicted probability of senility symptoms.

**Definition 3.10** (Conditional expectation function). The **conditional expectation function**  $E[Y \mid X]$  is the function (and hence random variable) of  $X$  obtained by evaluating  $E[Y \mid X = x]$  at  $X$ ; specifically,  $E[Y \mid X] = g(X)$  where  $g(x) \stackrel{\text{def}}{=} E[Y \mid X = x]$ .

Exm

**Example 3.8** (The conditional expectation function of the birthweight model). Continuing Example 3.5 and Example 3.6,  $(X, Y)$  (gestational age, birthweight) is modeled as bivariate normal. The general form of the conditional mean derived in Example 3.5, evaluated at an arbitrary  $x$  instead of just  $x = 40$ , gives the conditional expectation function directly:

$$\begin{aligned}
g(x) &= \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \\
&= -1484.9846 + 115.5283 x
\end{aligned}$$

which is exactly the fitted regression line (the general algebraic identity  $\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) = (\mu_Y - \rho \frac{\sigma_Y}{\sigma_X} \mu_X) + \rho \frac{\sigma_Y}{\sigma_X} x$  is what makes the bivariate-normal conditional mean linear in  $x$ ). As a check, evaluating at  $x = 40$  recovers Example 3.6's result:

$$g(40) = -1484.9846 + 115.5283 \times 40 = 3136.15 \text{ g}$$

**Exercise 3.1** (Expectation of a sum, given a joint PMF). Let  $(X, Y)$  be discrete with joint probability mass function:

|         | $Y = 0$ | $Y = 1$ |
|---------|---------|---------|
| $X = 0$ | 0.2     | 0.3     |
| $X = 1$ | 0.1     | 0.4     |

Compute  $E[X + Y]$ .

#### Solution

*Solution.* Treating  $(X, Y)$  as a single discrete random object taking one of the four values  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$ , LOTUS (Theorem 3.5) applied to  $g(x, y) = x + y$  gives directly:

$$\begin{aligned}
E[X + Y] &= \sum_{x \in \{0, 1\}} \sum_{y \in \{0, 1\}} (x + y) P(X = x, Y = y) \\
&= (0+0)(0.2) + (0+1)(0.3) + (1+0)(0.1) + (1+1)(0.4) \\
&= 0 + 0.3 + 0.1 + 0.8 \\
&= 1.2
\end{aligned}$$

As a check:  $E[X] = 0(0.5) + 1(0.5) = 0.5$  and  $E[Y] = 0(0.3) + 1(0.7) = 0.7$ , so  $E[X + Y] = E[X] + E[Y] = 1.2$ .

```

x_labs <- c("X=0", "X=0", "X=1", "X=1")
y_labs <- c("Y=0", "Y=1", "Y=0", "Y=1")
probs <- c(0.2, 0.3, 0.1, 0.4)

plotly::plot_ly(
  x = ~y_labs, y = ~probs, color = ~x_labs,
  colors = c("steelblue", "tomato"),
  type = "bar"
) |>
plotly::layout(
  barmode = "group",
  xaxis = list(title = "Y"),
  yaxis = list(title = "P(X = x, Y = y)", range = c(0, 0.5)),
  legend = list(title = list(text = "X value"))
)

```

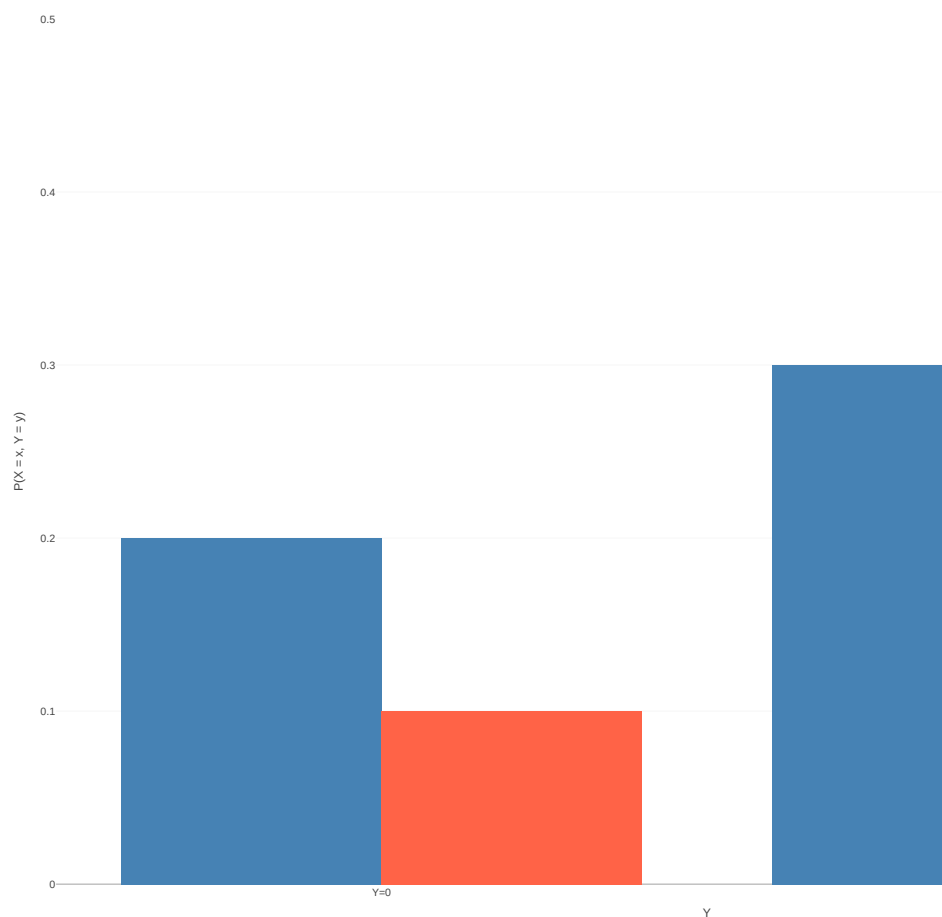


Figure 4: Joint probability mass function  $P(X = x, Y = y)$ . Marginal totals:  $P(X = 0) = 0.5$ ,  $P(X = 1) = 0.5$ ,  $P(Y = 0) = 0.3$ ,  $P(Y = 1) = 0.7$ .

Note that  $X$  and  $Y$  are **not** independent here:  $P(X = 0, Y = 0) = 0.2 \neq 0.15 = P(X = 0) P(Y = 0)$ . LOTUS applies regardless, since it requires only the *actual* joint mass function, not independence.

There are only four  $(x, y)$  pairs here, so summing them in any order — row by row, column by column, or any other listing — gives the same total by ordinary commutativity and associativity of addition; no result about interchanging summation order is needed. Fubini–Tonelli, covered next, is what’s needed once the support is countably infinite instead — see Exercise 3.2.

### 3.4 Fubini–Tonelli for expectations

The Riemann version of Fubini’s theorem<sup>4</sup>, stated in the math-prereqs chapter<sup>5</sup>, lets us switch the order of integration for continuous integrands on simple regions. For expectations against probability measures we use its measure-theoretic form<sup>6</sup>, which holds on any  $\sigma$ -finite measure space. The  $\sigma$ -finiteness condition is automatic for probability measures (every probability measure is finite, hence  $\sigma$ -finite), so Fubini–Tonelli<sup>7</sup> yields Corollary 3.1 directly.

**Corollary 3.1** (Joint-distribution form (without independence; corollary of Fubini–Tonelli)). *Let  $(X, Y)$  be jointly distributed random variables whose joint distribution has a density  $f_{X,Y}$  with respect to a product of  $\sigma$ -finite reference measures  $\mu_X \otimes \mu_Y$  on  $\mathcal{R}(X) \times \mathcal{R}(Y)$ , and let  $h : \mathcal{R}(X) \times \mathcal{R}(Y) \rightarrow \mathbb{R}$  be measurable. If either*

- (a)  $h(X, Y) \geq 0$  almost surely, or
- (b)  $E[|h(X, Y)|] < \infty$ ,

*then the expectation of  $h(X, Y)$  can be written as an iterated integral against  $f_{X,Y}$ , with the order of integration exchangeable:*

$$\begin{aligned} E[h(X, Y)] &= \int_{\mathcal{R}(X)} \left( \int_{\mathcal{R}(Y)} h(x, y) f_{X,Y}(x, y) d\mu_Y(y) \right) d\mu_X(x) \\ &= \int_{\mathcal{R}(Y)} \left( \int_{\mathcal{R}(X)} h(x, y) f_{X,Y}(x, y) d\mu_X(x) \right) d\mu_Y(y). \end{aligned}$$

The choice of reference measures covers three cases:

- **Both continuous:**  $\mu_X = \mu_Y = \text{Lebesgue measure}$ ;  $f_{X,Y}$  is the joint probability density function (PDF), and  $\int g(x) d\mu_X(x) = \int g(x) dx$ .
- **Both discrete:**  $\mu_X = \mu_Y = \text{counting measure}$ ;  $f_{X,Y}(x, y) = P(X = x, Y = y)$  is the joint probability mass function (PMF), and  $\int g(x) d\mu_X(x) = \sum_{x \in \mathcal{R}(X)} g(x)$ .
- **Mixed** (one continuous, one discrete): one reference measure is Lebesgue and the other is counting;  $f_{X,Y}(x, y) = f_{X|Y}(x | y) P(Y = y)$  (or  $P(X = x | Y = y) f_Y(y)$  if  $X$  is discrete and  $Y$  continuous), and the iterated integrals combine an ordinary integral with a sum. The conditional densities/PMFs here are defined the same way as in Definition 3.9, just conditioning on  $Y$  instead of  $X$ .

#### Proof

*Proof.* Apply Fubini–Tonelli<sup>a</sup> with  $\mu_1 = \mu_X$  and  $\mu_2 = \mu_Y$  to the integrand  $h(x, y) f_{X,Y}(x, y)$  on  $\mathcal{R}(X) \times \mathcal{R}(Y)$ . Lebesgue measure and counting measure on a countable set are each  $\sigma$ -finite, so  $\mu_X \otimes \mu_Y$  is  $\sigma$ -finite in all three cases. The relevant condition is (a) when  $h \geq 0$  and (b) when  $E[|h(X, Y)|] < \infty$ . Independence is not required. When  $X$  and  $Y$  are independent,  $f_{X,Y}(x, y) = f_X(x) f_Y(y)$  (or  $P(X = x, Y = y) = P(X = x) P(Y = y)$  in the discrete case), and the iterated integrals factor into separate integrals over the marginals.  $\square$

<sup>a</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)

<sup>4</sup>[math-prereqs.qmd#thm-fubini](#)

<sup>5</sup>[math-prereqs.qmd](#)

<sup>6</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)

<sup>7</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)

Exm

**Example 3.9** (Expectation of a product of independent variables). Let  $X \sim \text{Uniform}(0, 1)$  and  $Y \sim \text{Uniform}(0, 2)$ , independently distributed. Compute  $E[XY]$ .

We apply Corollary 3.1 (both-continuous case) with  $h(x, y) = xy$ . Since  $X$  and  $Y$  are independent with densities  $f_X(x) = 1$  on  $[0, 1]$  and  $f_Y(y) = \frac{1}{2}$  on  $[0, 2]$ , the joint density factors as  $f_{X,Y}(x, y) = f_X(x) f_Y(y) = \frac{1}{2}$ , and  $\mu_X = \mu_Y = \text{Lebesgue measure}$ :

$$\begin{aligned} E[XY] &= \int_0^1 \left( \int_0^2 xy \cdot \frac{1}{2} dy \right) dx \\ &= \int_0^1 x \left( \frac{1}{2} \int_0^2 y dy \right) dx \\ &= \int_0^1 x \cdot \frac{1}{2} \cdot \left[ \frac{y^2}{2} \right]_0^2 dx \\ &= \int_0^1 x \cdot \frac{1}{2} \cdot 2 dx \\ &= \int_0^1 x dx \\ &= \frac{1}{2} \end{aligned}$$

As a check:  $E[X] = \frac{1}{2}$ ,  $E[Y] = 1$ , and  $E[X] E[Y] = \frac{1}{2}$ .

Exm

**Example 3.10** (When independence fails: a counterexample). Correctly applying Corollary 3.1 requires the *actual* joint density  $f_{X,Y}$  — not the product of marginals  $f_X(x) f_Y(y)$ , which is valid only when  $X$  and  $Y$  are independent. Using the wrong joint density gives the wrong answer.

Let  $X \sim \text{Uniform}(0, 1)$  and set  $Y = X$  (so  $X$  and  $Y$  are perfectly correlated and **not** independent).

**True expectation:**

$$E[XY] = E[X \cdot X] = E[X^2] = \int_0^1 x^2 dx = \frac{1}{3}$$

**Erroneously applying the product-measure formula:**

Note that Fubini–Tonelli’s own conditions still hold here ( $h(x, y) = xy$  is nonnegative and integrable), so the error is not a failure of Fubini–Tonelli<sup>a</sup>. Rather, the error is using the *wrong measure*: the joint distribution of  $(X, X)$  is concentrated on the diagonal  $\{(x, x) : x \in [0, 1]\} \subset [0, 1]^2$ , which has Lebesgue measure zero in  $\mathbb{R}^2$ . The joint distribution is therefore **not** absolutely continuous with respect to two-dimensional Lebesgue measure, so **no joint density  $f_{X,Y}$  on  $[0, 1]^2$  exists**, which is the reference density Corollary 3.1 requires.

The following calculation is what someone would *erroneously* write if they assumed independence and used  $f_X(x) f_Y(y)$  as a “joint density” — a function that does not in fact correspond to the joint distribution of  $(X, X)$ . The marginals  $X \sim \text{Uniform}(0, 1)$  and  $Y \sim \text{Uniform}(0, 1)$  do have densities  $f_X = f_Y = 1$ , but the *product*  $f_X(x) f_Y(y) = 1$  on  $[0, 1]^2$  is the density of an *independent* pair, not of  $(X, X)$ :

$$\begin{aligned}
\int_0^1 \int_0^1 xy \cdot f_X(x) \cdot f_Y(y) \, dy \, dx &= \int_0^1 \int_0^1 xy \, dy \, dx \\
&= \int_0^1 x \left( \int_0^1 y \, dy \right) dx \\
&= \int_0^1 x \cdot \frac{1}{2} dx \\
&= \frac{1}{4}
\end{aligned}$$

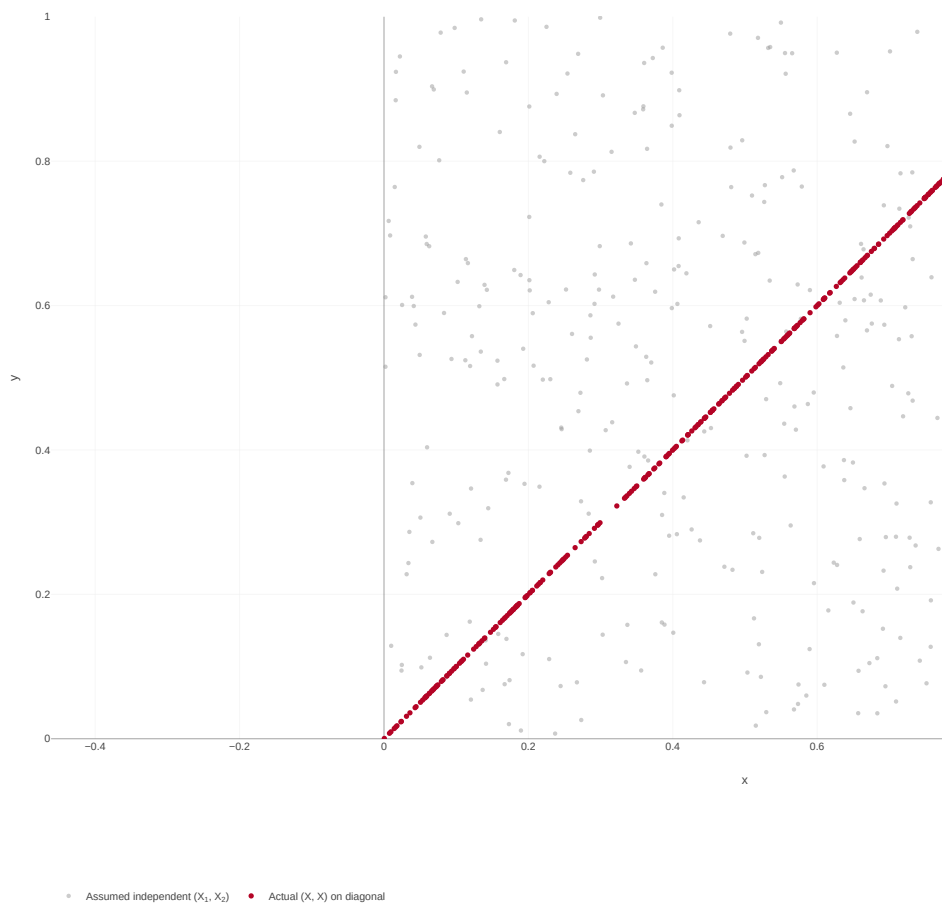
This calculation recovers  $E[XY]$  for *independent* uniforms ( $\frac{1}{4}$ ), not  $E[XX]$  for the perfectly correlated pair ( $\frac{1}{3}$ ). The lesson is that Corollary 3.1 requires the *actual* joint density  $f_{X,Y}$ . For independent  $(X, Y)$ , this factors as  $f_X(x) f_Y(y)$ ; for dependent  $(X, Y)$ ,  $f_{X,Y}$  need not factor — and for  $(X, X)$ , no joint density on  $\mathbb{R}^2$  exists at all, so Corollary 3.1 simply does not apply.

```

set.seed(204)
n <- 400
x_dep <- runif(n)
y_dep <- x_dep
x_ind <- runif(n)
y_ind <- runif(n)

plotly::plot_ly() |>
  plotly::add_trace(
    type = "scatter", mode = "markers",
    x = x_ind, y = y_ind,
    name = "Assumed independent (X<sub>1</sub>, X<sub>2</sub>)",
    marker = list(size = 5, color = "#999999", opacity = 0.5)
  ) |>
  plotly::add_trace(
    type = "scatter", mode = "markers",
    x = x_dep, y = y_dep,
    name = "Actual (X, X) on diagonal",
    marker = list(size = 6, color = "#b40426")
  ) |>
  plotly::layout(
    xaxis = list(title = "x", range = c(0, 1), scaleanchor = "y"),
    yaxis = list(title = "y", range = c(0, 1)),
    legend = list(orientation = "h", y = -0.2)
  )

```



Exm

**Example 3.11** (Both-continuous case: joint PDF on a non-rectangular support). Let  $(X, Y)$  have joint density  $f_{X,Y}(x, y) = 2$  for  $0 \leq x \leq y \leq 1$  (and 0 otherwise). Compute  $E[X + Y]$ . By Corollary 3.1:

$$\begin{aligned} E[X + Y] &= \int_0^1 \int_0^y (x + y) \cdot 2 \, dx \, dy \\ &= 2 \int_0^1 \left[ \frac{x^2}{2} + xy \right]_{x=0}^{x=y} dy \\ &= 2 \int_0^1 \left( \frac{y^2}{2} + y^2 \right) dy \\ &= 2 \int_0^1 \frac{3y^2}{2} dy \\ &= 3 \int_0^1 y^2 dy \\ &= 3 \cdot \frac{1}{3} \\ &= 1 \end{aligned}$$

```

n_grid <- 51
x_seq <- seq(0, 1, length.out = n_grid)
y_seq <- seq(0, 1, length.out = n_grid)

z_mat <- outer(x_seq, y_seq, function(x, y) {
  z <- rep(2, length(x))
  z[x > y] <- NA
  z
})

plotly::plot_ly(x = ~x_seq, y = ~y_seq, z = ~t(z_mat)) |>
  plotly::add_surface(showscale = FALSE) |>
  plotly::layout(scene = list(
    xaxis = list(title = "x"),
    yaxis = list(title = "y"),
    zaxis = list(title = "f(x, y)", range = c(0, 2.5)),
    camera = list(eye = list(x = 1.6, y = -1.6, z = 0.8))
  ))

```

**Exercise 3.2** (Both-discrete case, infinite support: joint PMF). Let  $X$  and  $Y$  be independent, each Geometric on  $\mathcal{R}(X) = \mathcal{R}(Y) = \{0, 1, 2, \dots\}$  with  $P(X = x) = (1 - p)p^x$  for a fixed  $p \in (0, 1)$  ( $X$  counts the number of failures before the first success in a sequence of independent trials with success probability  $1 - p$ ; likewise for  $Y$ ). Unlike Exercise 3.1, the support here is countably infinite. The joint PMF is  $P(X = x, Y = y) = (1 - p)^2 p^{x+y}$ . Compute  $E[X + Y]$ .

### Solution

*Solution.* Compute  $E[X + Y]$  using Corollary 3.1 with  $\mu_X = \mu_Y =$  counting measure and  $h(x, y) = x + y$ . Since  $h(x, y) = x + y \geq 0$  for every  $(x, y)$  in this support, condition (a) holds, so Corollary 3.1 (via Tonelli's theorem) guarantees the order of this now-infinite double sum is exchangeable — unlike the finite case, elementary algebra alone could not establish this. By Corollary 3.1 (both-discrete case), summing over  $y$  first for each fixed  $x$ :

$$\begin{aligned}
 E[X + Y] &= \sum_{x=0}^{\infty} \sum_{y=0}^{\infty} (x + y) P(X = x, Y = y) \\
 &= \sum_{x=0}^{\infty} \sum_{y=0}^{\infty} (x + y) (1 - p)^2 p^{x+y} \\
 &= \sum_{x=0}^{\infty} (1 - p)^2 p^x \sum_{y=0}^{\infty} (x + y) p^y \\
 &= \sum_{x=0}^{\infty} (1 - p)^2 p^x \left( x \sum_{y=0}^{\infty} p^y + \sum_{y=0}^{\infty} y p^y \right) \\
 &= \sum_{x=0}^{\infty} (1 - p)^2 p^x \left( \frac{x}{1 - p} + \frac{p}{(1 - p)^2} \right) \\
 &= \sum_{x=0}^{\infty} p^x [x(1 - p) + p] \\
 &= (1 - p) \sum_{x=0}^{\infty} x p^x + p \sum_{x=0}^{\infty} p^x \\
 &= (1 - p) \cdot \frac{p}{(1 - p)^2} + p \cdot \frac{1}{1 - p} \\
 &= \frac{p}{1 - p} + \frac{p}{1 - p} \\
 &= \frac{2p}{1 - p}
 \end{aligned}$$

using the standard geometric-series facts  $\sum_{y=0}^{\infty} p^y = \frac{1}{1 - p}$  and  $\sum_{y=0}^{\infty} y p^y = \frac{p}{(1 - p)^2}$  (e.g. Casella and Berger (2002)).

As a check,  $E[X] = E[Y] = \frac{p}{1 - p}$  (the mean of this Geometric distribution; Casella and Berger (2002)), so  $E[X + Y] = E[X] + E[Y] = \frac{2p}{1 - p}$  by linearity, matching.

```

p <- 0.4
exact_sum <- 2 * p / (1 - p)

n_trunc <- 500
support <- 0:n_trunc
joint_pmf_fn <- function(x, y) (1 - p)^2 * p^(x + y)
joint_probs <- outer(support, support, joint_pmf_fn)
h_vals <- outer(support, support, function(x, y) x + y)

sum_y_first <- sum(rowSums(h_vals * joint_probs))
sum_x_first <- sum(colSums(h_vals * joint_probs))

pander::pander(data.frame(
  quantity = c("exact (closed form)", "sum over y first, then x",
               "sum over x first, then y"),
  value = round(c(exact_sum, sum_y_first, sum_x_first), 6)
))

```

| quantity                 | value |
|--------------------------|-------|
| exact (closed form)      | 1.333 |
| sum over y first, then x | 1.333 |
| sum over x first, then y | 1.333 |

With  $p = 0.4$ , summing  $y$  first then  $x$ , and summing  $x$  first then  $y$ , agree with each other and with the closed form  $\frac{2p}{1-p} = 1.3333$  up to truncation error — exactly Tonelli’s guarantee.

**i** Tonelli’s nonnegativity condition, and what happens without it

The calculation in Exercise 3.2 only needed condition (a),  $h(X, Y) \geq 0$ , because  $h(x, y) = x + y$  is nonnegative on this support. For a **signed**  $h$ , interchanging an infinite double sum is not automatically valid — Corollary 3.1’s condition (b),  $E[|h(X, Y)|] < \infty$ , is what licenses it in that case. Without either condition, the two orders can genuinely disagree. A standard example (a signed array, not a probability distribution; see e.g. Rudin (1976) for the general theory of rearranging series): let  $a_{m,n} = 1$  if  $m = n$ ,  $a_{m,n} = -1$  if  $m = n + 1$ , and  $a_{m,n} = 0$  otherwise, for  $m, n = 0, 1, 2, \dots$ . Summing each row  $m$  first: row 0 has only the term  $a_{0,0} = 1$  (there is no valid  $n = -1$ ), so its row sum is 1; every row  $m \geq 1$  has  $a_{m,m} = 1$  and  $a_{m,m-1} = -1$ , so its row sum is 0. Summing the rows then gives  $1 + 0 + 0 + \dots = 1$ . Summing each column  $n$  first: every column  $n \geq 0$  has  $a_{n,n} = 1$  and  $a_{n+1,n} = -1$ , so its column sum is always 0, and summing the columns then gives  $0 + 0 + \dots = 0$ . The two orders give 1 and 0: genuinely different answers, confirming that a condition like (a) or (b) really is needed once the terms are no longer all nonnegative.

**Exercise 3.3** (Mixed case: one continuous variable, one discrete variable). Let  $Y \sim \text{Bernoulli}(0.6)$  and, given  $Y = y$ , let  $X | Y = y \sim \text{Uniform}(0, y + 1)$ . Compute  $E[X]$ .

**Solution**

*Solution.* Compute  $E[X]$  using Corollary 3.1 with  $\mu_X =$  Lebesgue measure,  $\mu_Y =$  counting measure, and  $h(x, y) = x$ .

The joint density w.r.t. Lebesgue  $\times$  counting measure is  $f_{X,Y}(x, y) = f_{X|Y}(x | y) P(Y = y)$ :

$$\begin{aligned}f_{X,Y}(x, 0) &= 1 \cdot 0.4 = 0.4 && \text{for } x \in [0, 1]; \\f_{X,Y}(x, 1) &= \tfrac{1}{2} \cdot 0.6 = 0.3 && \text{for } x \in [0, 2].\end{aligned}$$

By Corollary 3.1 (mixed case):

$$\begin{aligned}\mathbb{E}[X] &= \sum_{y \in \{0,1\}} \int_0^{y+1} x f_{X,Y}(x, y) dx \\&= \int_0^1 x \cdot 0.4 dx + \int_0^2 x \cdot 0.3 dx \\&= 0.4 \cdot \frac{1}{2} + 0.3 \cdot 2 \\&= 0.2 + 0.6 = 0.8\end{aligned}$$

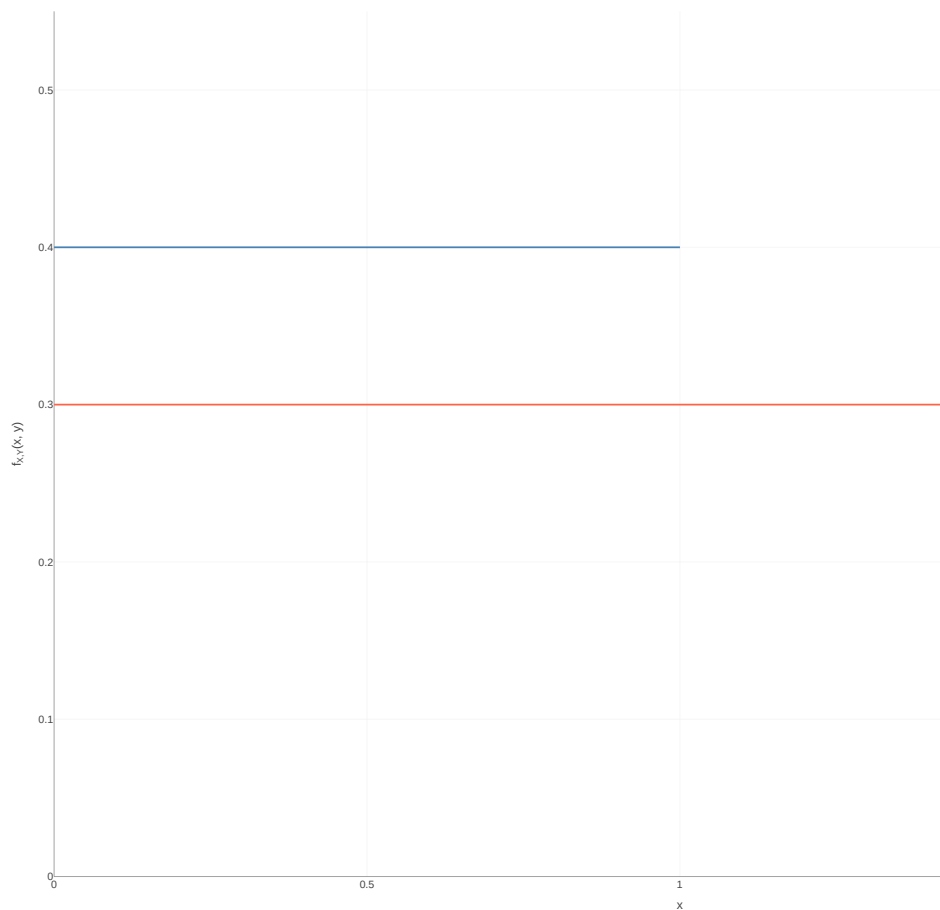
As a check using the law of total expectation:  $\mathbb{E}[X \mid Y = 0] = \frac{1}{2}$  and  $\mathbb{E}[X \mid Y = 1] = 1$ , so  $\mathbb{E}[X] = \frac{1}{2}(0.4) + 1(0.6) = 0.2 + 0.6 = 0.8$ .

```

x_fine <- seq(0, 2, by = 0.005)
df <- data.frame(
  x = c(x_fine[x_fine <= 1], x_fine),
  density = c(rep(0.4, sum(x_fine <= 1)), rep(0.3, length(x_fine))),
  label = c(
    rep("Y = 0 (P = 0.4)", sum(x_fine <= 1)),
    rep("Y = 1 (P = 0.6)", length(x_fine))
  )
)

plotly::plot_ly(
  df, x = ~x, y = ~density, color = ~label,
  colors = c("steelblue", "tomato")
) |>
plotly::add_lines() |>
plotly::layout(
  xaxis = list(title = "x"),
  yaxis = list(title = "fX,Y(x, y)", range = c(0, 0.55)),
  legend = list(title = list(text = "Y value"))
)

```



$Y$  takes only finitely many values here (two), so  $\sum_{y \in \{0,1\}} \int_0^{y+1} x f_{X,Y}(x, y) dx$  is just linearity of the integral applied to a two-term sum —  $\int g + \int k = \int(g + k)$  — not a genuine interchange of summation and integration order. Exercise 3.4 gives  $Y$  a countably infinite range instead, so the sum-of-integrals is genuinely infinite and Corollary 3.1's guarantee is load-bearing.

**Exercise 3.4** (Mixed case, infinite discrete support). Let  $Y$  be Geometric on  $\{0, 1, 2, \dots\}$  with  $P(Y = y) = (1-q)q^y$  for a fixed  $q \in (0, 1)$  and, given  $Y = y$ , let  $X | Y = y \sim \text{Uniform}(0, y+1)$ . Unlike Exercise 3.3,  $Y$ 's range here is countably infinite. Compute  $E[X]$ .

#### Solution

*Solution.* Compute  $E[X]$  using Corollary 3.1 with  $\mu_X =$  Lebesgue measure,  $\mu_Y =$  counting measure, and  $h(x, y) = x$ .

The joint density w.r.t. Lebesgue  $\times$  counting measure is  $f_{X,Y}(x, y) = f_{X|Y}(x | y) P(Y = y) = \frac{(1-q)q^y}{y+1}$  for  $x \in [0, y+1]$ .

Since  $h(x, y) = x \geq 0$  on this support, condition (a) holds, so Corollary 3.1 (via Tonelli's theorem) guarantees the now-infinite sum-of-integrals expression is valid — unlike in Exercise 3.3, this Fubini–Tonelli justification is required because the sum is infinite rather than finite.

By Corollary 3.1 (mixed case):

$$\begin{aligned}
 E[X] &= \sum_{y=0}^{\infty} \int_0^{y+1} x f_{X,Y}(x, y) dx \\
 &= \sum_{y=0}^{\infty} \frac{(1-q)q^y}{y+1} \int_0^{y+1} x dx \\
 &= \sum_{y=0}^{\infty} \frac{(1-q)q^y}{y+1} \cdot \frac{(y+1)^2}{2} \\
 &= \frac{1-q}{2} \sum_{y=0}^{\infty} (y+1) q^y \\
 &= \frac{1-q}{2} \left( \sum_{y=0}^{\infty} y q^y + \sum_{y=0}^{\infty} q^y \right) \\
 &= \frac{1-q}{2} \left( \frac{q}{(1-q)^2} + \frac{1}{1-q} \right) \\
 &= \frac{1-q}{2} \cdot \frac{q + (1-q)}{(1-q)^2} \\
 &= \frac{1-q}{2} \cdot \frac{1}{(1-q)^2} \\
 &= \frac{1}{2(1-q)}
 \end{aligned}$$

using the same geometric-series facts as Exercise 3.2 (e.g. Casella and Berger (2002)).

As a check using the law of total expectation:  $E[X | Y = y] = \frac{y+1}{2}$  (the mean of  $\text{Uniform}(0, y+1)$ ) and  $E[Y] = \frac{q}{1-q}$  (the mean of this Geometric distribution; Casella and Berger (2002)), so:

$$\begin{aligned}
E[X] &= E[E[X \mid Y]] \\
&= E\left[\frac{Y+1}{2}\right] \\
&= \frac{E[Y] + 1}{2} \\
&= \frac{1}{2} \left( \frac{q}{1-q} + 1 \right) \\
&= \frac{1}{2} \cdot \frac{q + (1-q)}{1-q} \\
&= \frac{1}{2(1-q)}
\end{aligned}$$

matching.

```

q <- 0.5
exact_ex <- 1 / (2 * (1 - q))

n_trunc <- 2000
y_vals <- 0:n_trunc
py_vals <- (1 - q) * q^y_vals
cond_mean <- (y_vals + 1) / 2
trunc_ex <- sum(cond_mean * py_vals)

pander::pander(data.frame(
  quantity = c("exact (closed form)", "truncated sum-of-integrals"),
  value = round(c(exact_ex, trunc_ex), 6)
))

```

| quantity                   | value |
|----------------------------|-------|
| exact (closed form)        | 1     |
| truncated sum-of-integrals | 1     |

With  $q = 0.5$ , the truncated sum-of-integrals matches the closed form  $\frac{1}{2(1-q)} = 1$ .

**Theorem 3.6** (Law of iterated expectations). *For any two random variables  $X$  and  $Y$ :*

$$E[Y] = E[E[Y \mid X]]$$

*Alternate names for this identity include: the **tower rule**, the **tower property**, the **law of total expectation**, and the **smoothing theorem**.*

#### **i** Proof

*Proof. Discrete case.* When  $X$  and  $Y$  are discrete, applying Definition 3.5 to  $E[E[Y \mid X]]$  and then the law of total probability (Theorem 1.4) applied to the countable partition  $\{X = x : x \in \mathcal{R}(X)\}$ :

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y \mid X]] &= \sum_{x \in \mathcal{R}(X)} \mathbb{E}[Y \mid X = x] \cdot \mathbb{P}(X = x) && \text{(definition of expectation of } \mathbb{E}[Y \mid X]) \\
&= \sum_{x \in \mathcal{R}(X)} \left( \sum_{y \in \mathcal{R}(Y)} y \cdot \mathbb{P}(Y = y \mid X = x) \right) \cdot \mathbb{P}(X = x) && \text{(definition of conditional expectation } \mathbb{E}[Y \mid X = x]) \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \sum_{x \in \mathcal{R}(X)} \mathbb{P}(Y = y \mid X = x) \cdot \mathbb{P}(X = x) && \text{(exchange order of summation)} \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \mathbb{P}(Y = y) && \text{(law of total probability over } X) \\
&= \mathbb{E}[Y] && \text{(definition of expectation of } Y)
\end{aligned}$$

**Continuous case.** When  $X$  and  $Y$  are continuous, applying Definition 3.5 to  $\mathbb{E}[\mathbb{E}[Y \mid X]]$  and then using Definition 3.8 for  $\mathbb{E}[Y \mid X = x]$ :

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y \mid X]] &= \int_{x \in \mathcal{R}(X)} \mathbb{E}[Y \mid X = x] \cdot p(X = x) dx \\
&= \int_{x \in \mathcal{R}(X)} \left( \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y \mid X = x) dy \right) \cdot p(X = x) dx \\
&= \int_{y \in \mathcal{R}(Y)} y \cdot \left( \int_{x \in \mathcal{R}(X)} p(Y = y \mid X = x) \cdot p(X = x) dx \right) dy \\
&= \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y) dy \\
&= \mathbb{E}[Y]
\end{aligned}$$

where the third equality exchanges the order of integration by condition (b) of Fubini–Tonelli<sup>a</sup> (the absolute-integrability case, **Fubini’s theorem**); this theorem requires  $\mathbb{E}[|Y|] < \infty$ , which is implicit in  $\mathbb{E}[Y]$  being defined, and the fourth equality uses  $\int_x p(Y = y \mid X = x) \cdot p(X = x) dx = \int_x p(X = x, Y = y) dx = p(Y = y)$  (marginalization of the joint density).  $\square$

<sup>a</sup>[math-prereqs.qmd#thm-fubini-tonelli](#)

**Theorem 3.7** (Conditional law of iterated expectations). *For random variables  $X$ ,  $Y$ , and  $Z$ :*

$$\mathbb{E}[Y \mid Z] = \mathbb{E}[\mathbb{E}[Y \mid X, Z] \mid Z]$$

*This identity is the tower rule applied conditionally on  $Z$ .*

### **i** Proof

*Proof.* For each fixed value  $z$  with positive probability or density:

**Discrete case.** Conditioning on  $Z = z$ , and applying the law of total probability to the partition  $\{X = x : x \in \mathcal{R}(X)\}$  under the conditional distribution given  $Z = z$ :

$$\begin{aligned}
E[E[Y | X, Z] | Z = z] &= \sum_{x \in \mathcal{R}(X)} E[Y | X = x, Z = z] \cdot P(X = x | Z = z) && \text{(definition of expectation)} \\
&= \sum_{x \in \mathcal{R}(X)} \left( \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y | X = x, Z = z) \right) \cdot P(X = x | Z = z) && \text{(definition of conditional expectation)} \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \sum_{x \in \mathcal{R}(X)} P(Y = y | X = x, Z = z) \cdot P(X = x | Z = z) && \text{(exchange order of summation)} \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y | Z = z) && \text{(law of total probability given } Z = z) \\
&= E[Y | Z = z] && \text{(definition of conditional expectation)}
\end{aligned}$$

**Continuous case.** Conditioning on  $Z = z$ , and integrating over  $X$  under the conditional density  $p(X = x | Z = z)$ :

$$\begin{aligned}
E[E[Y | X, Z] | Z = z] &= \int_{x \in \mathcal{R}(X)} E[Y | X = x, Z = z] \cdot p(X = x | Z = z) dx \\
&= E[Y | Z = z]
\end{aligned}$$

Therefore, as random variables of  $Z$ ,  $E[Y | Z] = E[E[Y | X, Z] | Z]$ . □

#### Exm

**Example 3.12** (Marginal expectation from conditional expectations). Suppose  $X$  is a binary random variable indicating treatment assignment ( $X = 1$  treated,  $X = 0$  control), with  $P(X = 1) = 0.5$ , and suppose the outcome  $Y$  has conditional expectations:

$$E[Y | X = 1] = 10, \quad E[Y | X = 0] = 6$$

By the law of iterated expectations (Theorem 3.6):

$$\begin{aligned}
E[Y] &= E[E[Y | X]] \\
&= E[Y | X = 1] \cdot P(X = 1) + E[Y | X = 0] \cdot P(X = 0) \\
&= 10 \cdot 0.5 + 6 \cdot 0.5 \\
&= 5 + 3 \\
&= 8
\end{aligned}$$

**Definition 3.11** (Expectation of a random matrix). For a random matrix  $\mathbf{A}$  of size  $m \times n$  with  $(i, j)$ -th element  $A_{ij}$ , the **expectation**  $E \mathbf{A}$  is the  $m \times n$  matrix whose  $(i, j)$ -th element is  $E[A_{ij}]$ :

$$E \mathbf{A} \stackrel{\text{def}}{=} \begin{pmatrix} E[A_{11}] & E[A_{12}] & \cdots & E[A_{1n}] \\ E[A_{21}] & E[A_{22}] & \cdots & E[A_{2n}] \\ \vdots & \vdots & \ddots & \vdots \\ E[A_{m1}] & E[A_{m2}] & \cdots & E[A_{mn}] \end{pmatrix}$$

In other words, expectation is applied **element-wise** to a random matrix.

### 3.5 Deviation, error, and noise

**Definition 3.12** (Deviation). A **deviation** is the difference between a value and a reference value. For any quantity  $z$  and reference value  $r$ :

$$z - r$$

In probability and statistics, “deviation” often means deviation from a [population mean](#). For a random variable  $Y$ :

$$Y - E[Y]$$

See: Wikipedia: Deviation (statistics)<sup>a</sup>

<sup>a</sup>[https://en.wikipedia.org/wiki/Deviation\\_\(statistics\)](https://en.wikipedia.org/wiki/Deviation_(statistics))

**Definition 3.13** (Deviation from a population or subpopulation mean). In probabilistic models, we call this quantity a **deviation from a mean**. It is often also called an **error** or **noise term** in other sources. For the random variable  $Y$ , define the [deviation](#) from its mean as:

$$e(Y) \stackrel{\text{def}}{=} Y - E[Y]$$

For a realized observation  $y$ :

$$e(y) \stackrel{\text{def}}{=} y - E[Y]$$

In regression settings, the reference mean is often conditional on covariates:  $e(y_i) \stackrel{\text{def}}{=} y_i - E[Y_i | X_i]$ .

In this course, we prefer “deviation” for this mean-deviation quantity. The terms “error” and “noise” are common aliases. We use “residual” (defined in the Linear regression chapter<sup>a</sup>) for deviations from fitted values. For notation in this course, we use  $e(\cdot)$  for these model/data deviations, and reserve  $\varepsilon(\cdot)$  for estimator-to-estimand deviations (see Estimation<sup>b</sup>).

See:

- Wikipedia: Errors and residuals<sup>c</sup>
- Wikipedia: Deviation (statistics)<sup>d</sup>
- Wikipedia: Linear regression — Notation and terminology<sup>e</sup>

<sup>a</sup>[Linear-models-overview.qmd#def-resid-fitted](#)

<sup>b</sup>[estimation.qmd#def-estimation-error](#)

<sup>c</sup>[https://en.wikipedia.org/wiki/Errors\\_and\\_residuals](https://en.wikipedia.org/wiki/Errors_and_residuals)

<sup>d</sup>[https://en.wikipedia.org/wiki/Deviation\\_\(statistics\)](https://en.wikipedia.org/wiki/Deviation_(statistics))

<sup>e</sup>[https://en.wikipedia.org/wiki/Linear\\_regression#Notation\\_and\\_terminology](https://en.wikipedia.org/wiki/Linear_regression#Notation_and_terminology)

### 3.6 Variance and related characteristics

**Definition 3.14** (Variance). The variance of a random variable  $X$  is the [expectation](#) of the squared [deviation from the mean](#); that is:

$$\text{Var}(X) \stackrel{\text{def}}{=} E[[e(X)]^2]$$

**Theorem 3.8** (Variance as expected squared deviation from the mean).

$$\text{Var}(X) = E[(X - E[X])^2]$$

**Proof**

*Proof.* Substituting the definition of  $e(X)$  from Definition 3.13 into Definition 3.14:

$$\text{Var}(X) \stackrel{\text{def}}{=} E[[e(X)]^2] = E[(X - E[X])^2].$$

□

**Theorem 3.9** (Simplified expression for variance).

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

**Proof**

*Proof.* By linearity of expectation, we have:

$$\begin{aligned} \text{Var}(X) &\stackrel{\text{def}}{=} \mathbb{E}[(e(X))^2] && \text{(definition of variance)} \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] && \text{(definition of deviation from mean)} \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] && \text{(expand binomial square)} \\ &= \mathbb{E}[X^2] - \mathbb{E}[2X\mathbb{E}[X]] + \mathbb{E}[(\mathbb{E}[X])^2] && \text{(linearity of expectation)} \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 && \text{(constants factor out of expectation)} \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 && \text{(algebraic simplification)} \end{aligned}$$

□

**Theorem 3.10** (Law of total variance). *For random variables  $X$  and  $Y$ :*

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])$$

where  $\text{Var}(Y | X) \stackrel{\text{def}}{=} \mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X]$ .

*Alternate names include: the **conditional variance formula**, **Eve's law**, and the **variance decomposition formula**.*

### i Proof

*Proof.* Write  $Y - \mathbb{E}[Y] = (Y - \mathbb{E}[Y | X]) + (\mathbb{E}[Y | X] - \mathbb{E}[Y])$ . Then:

$$(Y - \mathbb{E}[Y])^2 = (Y - \mathbb{E}[Y | X])^2 + (\mathbb{E}[Y | X] - \mathbb{E}[Y])^2 + 2(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y])$$

Taking expectation:

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - \mathbb{E}[Y])^2] \\ &\quad + 2\mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y])] \end{aligned}$$

For the cross-term:

**Discrete case.**

$$\begin{aligned} \mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y])] &= \sum_{x \in \mathcal{R}(X)} \mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y]) | X = x] \cdot \mathbb{P}(X = x) \\ &= \sum_{x \in \mathcal{R}(X)} (\mathbb{E}[Y | X = x] - \mathbb{E}[Y]) \cdot \mathbb{E}[Y - \mathbb{E}[Y | X = x] | X = x] \cdot \mathbb{P}(X = x) \\ &= 0 \end{aligned}$$

**Continuous case.**

$$\begin{aligned}
E[(Y - E[Y | X])(E[Y | X] - E[Y])] &= \int_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Y | X] - E[Y]) | X = x] \cdot p(X = x) dx \\
&= \int_{x \in \mathcal{R}(X)} (E[Y | X = x] - E[Y]) \cdot E[Y - E[Y | X = x] | X = x] \cdot p(X = x) dx \\
&= 0
\end{aligned}$$

Therefore:

$$\begin{aligned}
\text{Var}(Y) &= E[(Y - E[Y | X])^2] + E[(E[Y | X] - E[Y])^2] \\
&= E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])
\end{aligned}$$

□

**Definition 3.15** (Precision). The **precision** of a random variable  $X$ , often denoted  $\tau(X)$ ,  $\tau_X$ , or shorthand as  $\tau$ , is the inverse of that random variable's **variance**; that is:

$$\tau(X) \stackrel{\text{def}}{=} (\text{Var}(X))^{-1}$$

**Definition 3.16** (Standard deviation). The standard deviation of a random variable  $X$  is the square-root of the **variance** of  $X$ :

$$\text{SD}(X) \stackrel{\text{def}}{=} \sqrt{\text{Var}(X)}$$

**Definition 3.17** (Covariance). For any two one-dimensional random variables,  $X, Y$ :

$$\text{Cov}(X, Y) \stackrel{\text{def}}{=} E[(X - E[X])(Y - E[Y])]$$

**Theorem 3.11** (Alternative formula for covariance).

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

**Theorem 3.12** (Law of total covariance). For random variables  $X, Y$ , and  $Z$ :

$$\text{Cov}(Y, Z) = E[\text{Cov}(Y, Z | X)] + \text{Cov}(E[Y | X], E[Z | X])$$

where  $\text{Cov}(Y, Z | X) \stackrel{\text{def}}{=} E[(Y - E[Y | X])(Z - E[Z | X]) | X]$ .

Alternate names include: the **covariance decomposition formula** and the **conditional covariance formula**.

### **i** Proof

*Proof.* Write:

$$\begin{aligned}
Y - E[Y] &= (Y - E[Y | X]) + (E[Y | X] - E[Y]) \\
Z - E[Z] &= (Z - E[Z | X]) + (E[Z | X] - E[Z])
\end{aligned}$$

Then:

$$\begin{aligned}
\text{Cov}(Y, Z) &= E[(Y - E[Y])(Z - E[Z])] \\
&= E[(Y - E[Y | X])(Z - E[Z | X])] \\
&\quad + E[(Y - E[Y | X])(E[Z | X] - E[Z])] \\
&\quad + E[(E[Y | X] - E[Y])(Z - E[Z | X])] \\
&\quad + E[(E[Y | X] - E[Y])(E[Z | X] - E[Z])]
\end{aligned}$$

For the two mixed terms:

**Discrete case.**

$$\begin{aligned}
E[(Y - E[Y | X])(E[Z | X] - E[Z])] &= \sum_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Z | X] - E[Z]) | X = x] \cdot P(X = x) \\
&= \sum_{x \in \mathcal{R}(X)} (E[Z | X = x] - E[Z]) \cdot E[Y - E[Y | X = x] | X = x] \cdot P(X = x) \\
&= 0
\end{aligned}$$

and similarly:

$$E[(E[Y | X] - E[Y])(Z - E[Z | X])] = 0.$$

**Continuous case.**

$$\begin{aligned}
E[(Y - E[Y | X])(E[Z | X] - E[Z])] &= \int_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Z | X] - E[Z]) | X = x] \cdot p(X = x) dx \\
&= \int_{x \in \mathcal{R}(X)} (E[Z | X = x] - E[Z]) \cdot E[Y - E[Y | X = x] | X = x] \cdot p(X = x) dx \\
&= 0
\end{aligned}$$

and similarly:

$$E[(E[Y | X] - E[Y])(Z - E[Z | X])] = 0.$$

Hence:

$$\begin{aligned}
\text{Cov}(Y, Z) &= E[(Y - E[Y | X])(Z - E[Z | X])] + E[(E[Y | X] - E[Y])(E[Z | X] - E[Z])] \\
&= E[\text{Cov}(Y, Z | X)] + \text{Cov}(E[Y | X], E[Z | X])
\end{aligned}$$

□

---

**Lemma 3.1** (The covariance of a variable with itself is its variance). *For any random variable  $X$ :*

$$\text{Cov}(X, X) = \text{Var}(X)$$

**i** Proof

*Proof.*

$$\begin{aligned}
\text{Cov}(X, X) &= E[XX] - E[X]E[X] \\
&= E[X^2] - (E[X])^2 \\
&= \text{Var}(X)
\end{aligned}$$

□

**Definition 3.18** (Variance/covariance of a  $p \times 1$  random vector). For a  $p \times 1$  dimensional random vector  $\tilde{X}$ ,

$$\begin{aligned}\text{Var}(\tilde{X}) &\stackrel{\text{def}}{=} \text{Cov}(\tilde{X}) \\ &\stackrel{\text{def}}{=} \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top]\end{aligned}$$

**Theorem 3.13** (Elements of the variance-covariance matrix are pairwise covariances). For a  $p \times 1$  random vector  $\tilde{X} = (X_1, \dots, X_p)^\top$ , the  $(i, j)$ -th element of  $\text{Var}(\tilde{X})$  is  $\text{Cov}(X_i, X_j)$ :

$$\text{Var}(\tilde{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{pmatrix}$$

#### Proof

*Proof.* Let  $\mu_i = \mathbb{E}[X_i]$  for  $i = 1, \dots, p$ , so  $\mathbb{E} \tilde{X} = (\mu_1, \dots, \mu_p)^\top$ . By Definition 3.18:

$$\begin{aligned}\text{Var}(\tilde{X}) &= \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top] \\ &= \mathbb{E} \left[ \begin{pmatrix} X_1 - \mu_1 \\ \vdots \\ X_p - \mu_p \end{pmatrix} (X_1 - \mu_1 \quad \cdots \quad X_p - \mu_p) \right] \\ &= \mathbb{E} \left[ \begin{pmatrix} (X_1 - \mu_1)(X_1 - \mu_1) & \cdots & (X_1 - \mu_1)(X_p - \mu_p) \\ \vdots & \ddots & \vdots \\ (X_p - \mu_p)(X_1 - \mu_1) & \cdots & (X_p - \mu_p)(X_p - \mu_p) \end{pmatrix} \right] \\ &= \begin{pmatrix} \mathbb{E}[(X_1 - \mu_1)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_1 - \mu_1)(X_p - \mu_p)] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[(X_p - \mu_p)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_p - \mu_p)(X_p - \mu_p)] \end{pmatrix} \\ &= \begin{pmatrix} \text{Cov}(X_1, X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Cov}(X_p, X_p) \end{pmatrix} \\ &= \begin{pmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Var}(X_p) \end{pmatrix}\end{aligned}$$

where:

- the step from the third to fourth line uses Definition 3.11,
- the step from the fourth to fifth line uses Definition 3.17, and
- the last step uses Lemma 3.1.

□

**Theorem 3.14** (Alternate expression for variance of a random vector).

$$\text{Var}(\tilde{X}) = \mathbb{E}[\tilde{X} \tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top$$

---

**i** Proof

*Proof.*

$$\begin{aligned}\text{Var}(\tilde{X}) &= \mathbb{E}\left[(\tilde{X} - \mathbb{E}\tilde{X})(\tilde{X} - \mathbb{E}\tilde{X})^\top\right] \\ &= \mathbb{E}\left[\tilde{X}\tilde{X}^\top - \tilde{X}(\mathbb{E}\tilde{X})^\top - (\mathbb{E}\tilde{X})\tilde{X}^\top + (\mathbb{E}\tilde{X})(\mathbb{E}\tilde{X})^\top\right] \\ &= \mathbb{E}\left[\tilde{X}\tilde{X}^\top\right] - (\mathbb{E}\tilde{X})(\mathbb{E}\tilde{X})^\top - (\mathbb{E}\tilde{X})(\mathbb{E}\tilde{X})^\top + (\mathbb{E}\tilde{X})(\mathbb{E}\tilde{X})^\top \\ &= \mathbb{E}\left[\tilde{X}\tilde{X}^\top\right] - (\mathbb{E}\tilde{X})(\mathbb{E}\tilde{X})^\top\end{aligned}$$

□

---

**Theorem 3.15** (Variance of a linear combination). *For any vector of random variables  $\tilde{X} = (X_1, \dots, X_n)$  and corresponding vector of constants  $\tilde{a} = (a_1, \dots, a_n)$ , the variance of their linear combination is:*

$$\begin{aligned}\text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)\end{aligned}$$

---

**i** Proof

*Proof.* Left to the reader...

□

---

**Corollary 3.2** (Variance of a sum of two random variables). *For any two random variables  $X$  and  $Y$  and scalars  $a$  and  $b$ :*

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2(a \cdot b) \text{Cov}(X, Y)$$

---

**i** Proof

*Proof.* Apply Theorem 3.15 with  $n = 2$ ,  $X_1 = X$ , and  $X_2 = Y$ .  
Or, see <https://statproofbook.github.io/P/var-lincomb.html>

□

---

**Definition 3.19** (Homoskedastic and heteroskedastic). A random variable  $Y$  is **homoskedastic** (with respect to covariates  $X$ ) if the **variance** of  $Y$  does not vary with  $X$ :

$$\text{Var}(Y|X = x) = \sigma^2, \forall x$$

Otherwise it is **heteroskedastic**.

**Definition 3.20** (Statistical independence). A set of random variables  $X_1, \dots, X_n$  are **statistically independent** if their joint **probability** is equal to the product of their marginal **probabilities**:

$$\Pr(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(X_i = x_i)$$

 **Tip**

The symbol for independence,  $\perp$ , is essentially just  $\prod$  upside-down. So the symbol can remind you of its definition (Definition 3.20).

**Definition 3.21** (Conditional independence). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally statistically independent** given a set of covariates  $X_1, \dots, X_n$  if the joint **probability** of the  $Y_i$ s given the  $X_i$ s is equal to the product of their marginal **probabilities**:

$$\Pr(Y_1 = y_1, \dots, Y_n = y_n | X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(Y_i = y_i | X_i = x_i)$$

**Definition 3.22** (Identically distributed). A set of random variables  $X_1, \dots, X_n$  are **identically distributed** if they have the same range  $\mathcal{R}(X)$  and if their marginal distributions  $P(X_1 = x_1), \dots, P(X_n = x_n)$  are all equal to some shared distribution  $P(X = x)$ :

$$\forall i \in \{1 : n\}, \forall x \in \mathcal{R}(X) : P(X_i = x) = P(X = x)$$

**Definition 3.23** (Conditionally identically distributed). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally identically distributed** given a set of covariates  $X_1, \dots, X_n$  if  $Y_1, \dots, Y_n$  have the same range  $\mathcal{R}(X)$  and if the distributions  $P(Y_i = y_i | X_i = x_i)$  are all equal to the same distribution  $P(Y = y | X = x)$ :

$$P(Y_i = y | X_i = x) = P(Y = y | X = x)$$

**Definition 3.24** (Independent and identically distributed). A set of random variables  $X_1, \dots, X_n$  are **independent and identically distributed** (shorthand: “ $X_i$  iid”) if they are **statistically independent** and **identically distributed**.

**Definition 3.25** (Conditionally independent and identically distributed). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally independent and identically distributed** (shorthand: “ $Y_i | X_i$  ciid” or just “ $Y_i | X_i$  iid”) given a set of covariates  $X_1, \dots, X_n$  if  $Y_1, \dots, Y_n$  are **conditionally independent** given  $X_1, \dots, X_n$  and  $Y_1, \dots, Y_n$  are **conditionally identically distributed** given  $X_1, \dots, X_n$ .

### 3.7 The Central Limit Theorem

The sum of many independent or nearly-independent random variables with small variances (relative to the number of RVs being summed) produces bell-shaped distributions.

For example, consider the sum of five dice (Figure 8).

```
library(dplyr)
dist =
  expand.grid(1:6, 1:6, 1:6, 1:6, 1:6) |>
  rowwise() |>
```

```

mutate(total = sum(c_across(everything())) |>
ungroup() |>
count(total) |>
mutate(`p(X=x)` = n/sum(n))

library(ggplot2)

dist |>
  ggplot() +
  aes(x = total, y = `p(X=x)`) +
  geom_col() +
  xlab("sum of dice (x)") +
  ylab("Probability of outcome, Pr(X=x)") +
  expand_limits(y = 0)

```

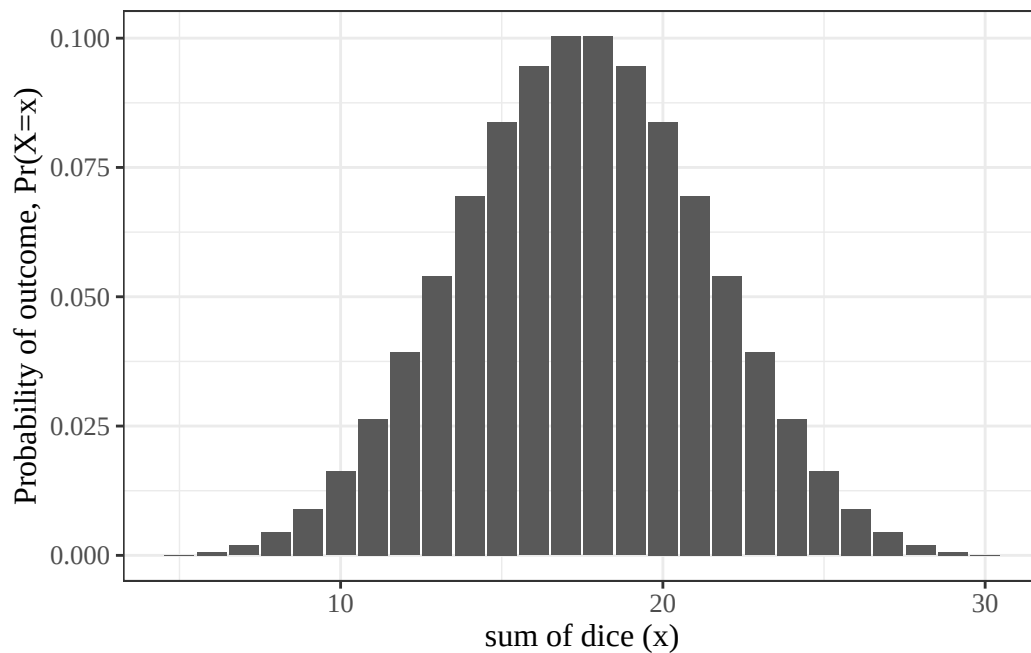


Figure 8: Distribution of the sum of five dice

In comparison, the outcome of just one die is not bell-shaped (Figure 9).

```

library(dplyr)
dist =
  expand.grid(1:6) |>
  rowwise() |>
  mutate(total = sum(c_across(everything())) |>
ungroup() |>
count(total) |>
mutate(`p(X=x)` = n/sum(n))

library(ggplot2)

dist |>
  ggplot() +

```

```

aes(x = total, y = `p(X=x)`) +
geom_col() +
xlab("sum of dice (x)") +
ylab("Probability of outcome, Pr(X=x)") +
expand_limits(y = 0)

```

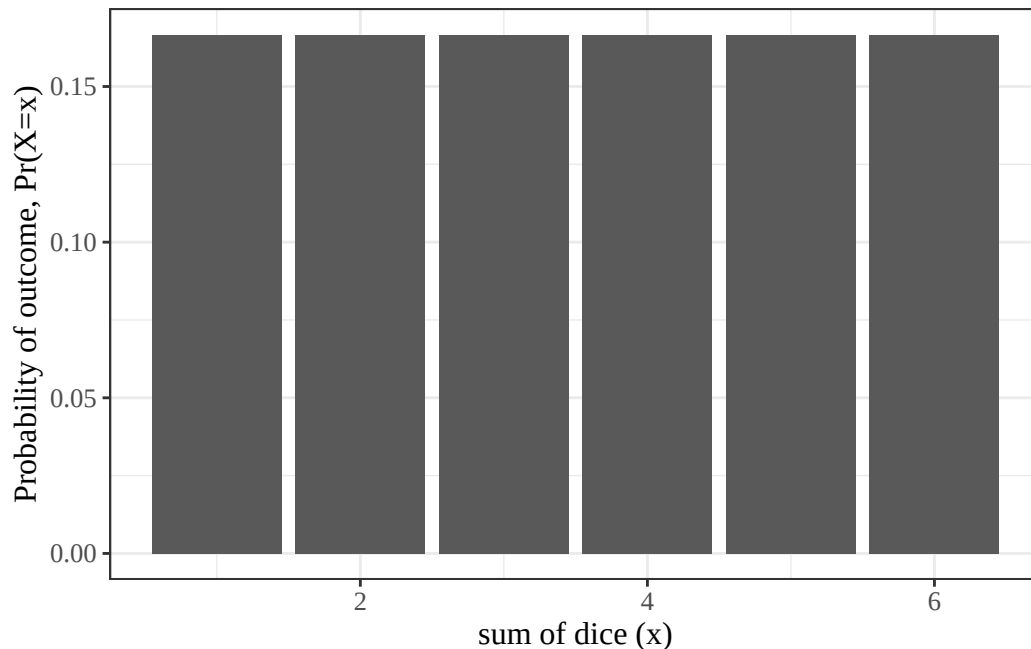


Figure 9: Distribution of the outcome of one die

What distribution does a single die have?

Answer: discrete uniform on 1:6.

## 4 Additional resources

- Miller (2017)

## References

- Billingsley, Patrick. 1995. *Probability and Measure*. 3rd ed. Wiley Series in Probability and Mathematical Statistics. Wiley.
- Casella, George, and Roger Berger. 2002. *Statistical Inference*. 2nd ed. Cengage Learning. <https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/>.
- Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.
- Gut, Allan. 2013. *Probability: A Graduate Course*. 2nd ed. Springer Texts in Statistics. Springer. <https://doi.org/10.1007/978-1-4614-4708-5>.
- Kalbfleisch, John D, and Ross L Prentice. 2011. *The Statistical Analysis of Failure Time Data*.

John Wiley & Sons.

- Klein, John P, and Melvin L Moeschberger. 2003. *Survival Analysis: Techniques for Censored and Truncated Data*. Vol. 1230. Springer. <https://link.springer.com/book/10.1007/b97377>.
- Kleinbaum, David G, and Mitchel Klein. 2012. *Survival Analysis: A Self-Learning Text*. 3rd ed. Springer. <https://link.springer.com/book/10.1007/978-1-4419-6646-9>.
- Miller, Steven J. 2017. *The Probability Lifesaver : All the Tools You Need to Understand Chance*. A Princeton Lifesaver Study Guide. Princeton University Press. <https://press.princeton.edu/books/hardcover/9780691149547/the-probability-lifesaver>.
- Rothman, Kenneth J., Timothy L. Lash, Tyler J. VanderWeele, and Sebastien Haneuse. 2021. *Modern Epidemiology*. Fourth edition. Wolters Kluwer.
- Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.
- Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.
- Wikipedia contributors. 2026. *Law of the Unconscious Statistician* — *Wikipedia, the Free Encyclopedia*. [https://en.wikipedia.org/wiki/Law\\_of\\_the\\_unconscious\\_statistician](https://en.wikipedia.org/wiki/Law_of_the_unconscious_statistician).